

# One Hundred Fifty-six Candidates for Datasets 50–200

Michael Volk

August 27, 2026

The database holds 49 schematized and L0–L4-verified *Saccharomyces cerevisiae* datasets. Reaching 200 needs 151 more. Table 1 ranks 156 candidates and Table 2 gives each one’s citation, link and data location; rows 1–151 are the recommended set and the remaining 5 are a ranked reserve. Table 5 records what was considered and dropped, which is where the one hard constraint shows up: a dataset earns a place only if the total genomic content of each strain can be reconstructed.

Section status:  ready  author-reviewed, keep stable  not done, agents may edit freely

## Contents

|          |                                                                                                                                                       |           |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>1</b> | <b>The selection rule</b>                                            | <b>2</b>  |
| 1.1      | Sequence reconstruction is the gate, not a preference                | 2         |
| 1.2      | Tier, and what rows are ranked on                                    | 2         |
| 1.3      | Where the cut falls, and what it cost                                | 3         |
| <b>2</b> | <b>The ranked list</b>                                               | <b>3</b>  |
| <b>3</b> | <b>What the recommended 151 add up to</b>                            | <b>29</b> |
| <b>4</b> | <b>The classes</b>                                                   | <b>29</b> |
| 4.1      | Natural isolates, and where sequence diversity crosses over         | 29        |
| 4.2      | Tolerance on recalcitrant biomass, and product tolerance           | 29        |
| 4.3      | CRISPR library screens                                             | 30        |
| 4.4      | Expression across conditions                                       | 30        |
| 4.5      | Metabolite and precursor coverage                                  | 30        |
| 4.6      | Regulatory DNA, deep scans and combinatorial genomes               | 31        |
| 4.7      | Perturbation modality and reference backbone                       | 31        |
| <b>5</b> | <b>Rows that bear on a yeast Perturb-seq</b>                       | <b>31</b> |
| 5.1      | The quadrant that matters is nearly empty                          | 33        |
| 5.2      | Depth rather than breadth                                          | 33        |
| 5.3      | The synergy question is answerable, and not yet answered           | 34        |
| <b>6</b> | <b>What was dropped</b>                                            | <b>34</b> |
| 6.1      | Four rows were not what their names said                           | 35        |
| 6.2      | Overlap is now checked by script, and the check is not yet strong  | 35        |
| 6.3      | The expansion tranche is less checked than the first 73            | 36        |
| <b>7</b> | <b>Before a loader is written</b>                                  | <b>36</b> |

# 1 The selection rule ✗

## 1.1 Sequence reconstruction is the gate, not a preference ✗

A phenotype is only trainable against a genotype that can be written down. Every row in Table 1 therefore names a **sequence basis**: the route by which the total genomic content of one strain would be reconstructed. The eleven bases in use are

- **S288C-KO** – the reference genome minus one cataloged ORF (BY4741, BY4742, BY4743). Thirty-six of the recommended 151 sit here, the largest single group.
- **S288C-KO/het** – a heterozygous diploid deletion, so a dosage edit rather than a null.
- **S288C+guide** – the genome is unedited and the perturbation is a designed cassette plus a guide target. Regulatory, not sequence-level.
- **S288C+designed-edit** – a designed single-nucleotide or small-indel library. Designed, and not verified per strain.
- **S288C+tag** – the reference plus a designed tag, degon or promoter cassette.
- **S288C+ORF-plasmid** – the reference plus a barcoded ORF on a known plasmid.
- **S288C+reporter-locus** – the reference plus a designed regulatory sequence driving a reporter, which is what every massively parallel reporter assay perturbs.
- **isolate-WGS** – a published assembly or variant call set per isolate.
- **segregant-WGS** – genotype calls or whole-genome sequence per progeny.
- **engineered-chassis** – a named production strain plus heterologous cassettes.
- **reference-only** – wild type, where the environment carries the perturbation.

What fails the gate is a strain whose genome is unknown even in principle. Genome shuffling and adaptive laboratory evolution without resequencing both land there: the selected progeny differ from their parents at unmapped positions, so there is no genotype to map a phenotype from. Those are dropped in Table 5 rather than ranked low, since no amount of phenotypic value substitutes for a missing genotype.

The **S288C+designed-edit** and **S288C+guide** bases carry a weaker claim than the others and the distinction is worth holding. A designed variant library states what each member was built to be, not what it turned out to be, so its genotypes are designed rather than realized. Ingesting one commits to storing the design and a verification status, not a verified sequence.

## 1.2 Tier, and what rows are ranked on ✗

Tier is assigned from stated bars rather than judgment:

- **Tier 1** clears all three of  $\geq 10^3$  genotypes,  $\geq 10^4$  instances, and a clean sequence basis, and lands in one of the nine classes.
- **Tier 2** clears two of the three, or is the only dataset covering a requested class, or directly de-risks the Perturb-seq proposal.
- **Tier 3** is a modality or reference backbone the substrate needs without itself carrying a large genotype axis.
- **Tier 4** is real but small, coarsely labeled, or expensive to extract.

The bars apply to what a row contains rather than to its reputation. Piotrowski 2017 has the largest compound library in yeast at 13,524 compounds and still fails Tier 1, because 157 strains is not a genotype axis. Turco 2023 has the widest environment axis at 7,536 and fails for a different reason: it is a meta-aggregation, so its instance count re-serves primary screens, several of them already built, and the net-new fraction is unknown until it is de-duplicated against the rest of this table.

Within a tier, rows sort by **measurements**, meaning instances times phenotype dimensionality. Instances alone is the wrong statistic and gets the order visibly wrong: Muenzner 2024 is 796 instances, which ranks it below a mid-sized fitness screen, but each instance is a proteome, so it is roughly  $1.6 \times 10^6$  numbers. This is the same normalization the supported-datasets table already applies through its gzip-signal column.

### 1.3 Where the cut falls, and what it cost ✗

The cut at 151 is arithmetic,  $200 - 49$ , not a quality threshold. Rows on both sides of it are close, so it should be read as a budget rather than a verdict.

A dataset named explicitly in the scoping request is pinned into the recommended set even if it ranks below the cut, and the pin is reported rather than folded into the score, so the set stays auditable. No pin is currently in force: at 151 rows the recommended set is wide enough that every requested dataset ranks inside it unaided.

Displacement, when a pin does bind, picks the weakest non-requested row by tier first and measurements second. The tier term is load-bearing rather than decorative: ranking on measurements alone selected Puddu 2019 for eviction, and that row is the whole-collection sequencing on which every **S288C-KO** sequence basis in this table rests.

## 2 The ranked list ✗

**Table 1:** Ranked candidates to take the database from 49 supported datasets to 200. *Genotypes* and *Env* are the perturbation and condition axes. *Instances* is the number of genotype $\times$ environment records, marked  $\dagger$  where it is the product of the two axes rather than a figure the paper reports, and  $\ddagger$  where it is an order-of-magnitude estimate. *Meas.* is instances times phenotype dimensionality, which is what a vector-valued panel actually contributes and what rows are ranked on. *Sequence basis* is how the total genomic content of one strain would be reconstructed; a row with no such route is excluded rather than ranked (Table 5). Tier is defined in Sec. 1. *Link* resolves to the source and is clickable; the full citation and the data location are in Table 2. A  $\bullet$  marks a row bearing on the Perturb-seq proposal (Table 4). A superscript **B** marks a row already attempted and blocked on data access, and **L** one that already has a loader in flight; neither is an untouched candidate. Rows 1–151 are the recommended set.

| #  | Dataset                                                                         | Class                    | Genotypes                    | Env                 | Instances                  | Meas.             | Phenotype                                         | Sequence basis       | Link                                                                                    |
|----|---------------------------------------------------------------------------------|--------------------------|------------------------------|---------------------|----------------------------|-------------------|---------------------------------------------------|----------------------|-----------------------------------------------------------------------------------------|
| 1  | <b>Boocock 2025 (single-cell eQTL mapping) <math>\bullet</math></b>             | Expression / single cell | 393 segregants, 27,744 cells | glucose / galactose | $5.5 \times 10^4 \dagger$  | $3.3 \times 10^8$ | single-cell transcriptome + eQTL                  | segregant-WGS        | <a href="https://doi.org/10.7554/eLife.95566">doi:10.7554/eLife.95566</a>               |
| 2  | <b>de Boer 2020 (100M random promoters) <math>\bullet</math></b>                | Regulatory DNA           | >100M random promoters       | complex + defined   | $1.0 \times 10^8$          | $1.0 \times 10^8$ | expression driven (YFP sort-seq)                  | S288C+reporter-locus | <a href="https://doi.org/10.1038/s41587-019-0315-8">doi:10.1038/s41587-019-0315-8</a>   |
| 3  | <b>Vaishnav 2022 (regulatory DNA fitness landscape) <math>\bullet</math></b>    | Regulatory DNA           | 2e7 promoter variants        | 1                   | $2.0 \times 10^7 \ddagger$ | $2.0 \times 10^7$ | expression + inferred fitness                     | S288C+reporter-locus | <a href="https://doi.org/10.1038/s41586-022-04506-6">doi:10.1038/s41586-022-04506-6</a> |
| 4  | <b>Lee 2014 (HIP-HOP fitness signatures) <sup>B</sup></b>                       | Tolerance / robustness   | 11,000 (het + hom)           | 3,356 compounds     | $1.3 \times 10^7$          | $1.3 \times 10^7$ | chemogenomic fitness score                        | S288C-KO             | <a href="https://doi.org/10.1126/science.1250217">doi:10.1126/science.1250217</a>       |
| 5  | <b>Albert 2018 (eQTL in 1,012 segregants) <math>\bullet</math></b>              | Natural variation        | 1,012 sequenced segregants   | 1                   | $1.0 \times 10^3$          | $5.8 \times 10^6$ | mRNA abundance (RNA-seq)                          | segregant-WGS        | <a href="https://doi.org/10.7554/eLife.35471">doi:10.7554/eLife.35471</a>               |
| 6  | <b>Jakobson 2025 (genome-to-proteome map)</b>                                   | Natural variation        | 800 F6 segregants            | 1                   | 800                        | $1.6 \times 10^6$ | protein abundance, >6,400 pQTL                    | segregant-WGS        | <a href="https://doi.org/10.1126/science.adu3198">doi:10.1126/science.adu3198</a>       |
| 7  | <b>Muenzner 2024 (natural-isolate proteome)</b>                                 | Natural variation        | 796 natural isolates         | 1                   | 796                        | $1.6 \times 10^6$ | protein abundance (DIA-MS)                        | isolate-WGS          | <a href="https://doi.org/10.1038/s41586-024-07442-9">doi:10.1038/s41586-024-07442-9</a> |
| 8  | <b>Hale 2024 (CRISPRi x natural variation) <math>\bullet</math></b>             | Natural variation        | 169 segregants x 1,721 genes | 2 induced           | $1.4 \times 10^6 \dagger$  | $1.4 \times 10^6$ | relative fitness (double-barcode seq)             | segregant-WGS        | <a href="https://doi.org/10.1038/s41467-024-48626-1">doi:10.1038/s41467-024-48626-1</a> |
| 9  | <b>Jackson 2020 (TF-deletion single-cell atlas) <math>\bullet</math></b>        | Expression / single cell | 72 TF deletions + wild type  | YPD / minimal       | 144 $\dagger$              | $8.6 \times 10^5$ | single-cell transcriptome                         | S288C-KO             | <a href="https://doi.org/10.7554/eLife.51254">doi:10.7554/eLife.51254</a>               |
| 10 | <b>Nguyen Ba 2022 (barcoded bulk QTL, 100k segregants) <math>\bullet</math></b> | Natural variation        | 100,000 barcoded segregants  | 8 conditions        | $8.0 \times 10^5 \dagger$  | $8.0 \times 10^5$ | fitness (Bar-seq)                                 | segregant-WGS        | <a href="https://doi.org/10.7554/eLife.73983">doi:10.7554/eLife.73983</a>               |
| 11 | <b>Galardini 2019 (four backgrounds x 38 conditions)</b>                        | Natural variation        | 3,786 KOs x 4 backgrounds    | 38 conditions       | $5.8 \times 10^5 \dagger$  | $5.8 \times 10^5$ | colony-size fitness (S-score)                     | isolate-WGS          | <a href="https://doi.org/10.15252/msb.20198831">doi:10.15252/msb.20198831</a>           |
| 12 | <b>Bloom 2019 (16-parent cross)</b>                                             | Natural variation        | 14,000 segregants            | 38 traits           | $5.3 \times 10^5 \dagger$  | $5.3 \times 10^5$ | quantitative growth traits                        | segregant-WGS        | <a href="https://doi.org/10.7554/eLife.49212">doi:10.7554/eLife.49212</a>               |
| 13 | <b>Parsons 2006 (bioactive-compound profiling)</b>                              | Tolerance / robustness   | 5,000 viable YKO             | 82 compounds        | $4.1 \times 10^5 \dagger$  | $4.1 \times 10^5$ | hypersensitivity score                            | S288C-KO             | <a href="https://doi.org/10.1016/j.cell.2006.06.040">doi:10.1016/j.cell.2006.06.040</a> |
| 14 | <b>Dutta 2026 (barcoded natural-isolate chemical response)</b>                  | Natural variation        | 520 natural isolates         | >600 compounds      | $3.1 \times 10^5 \dagger$  | $3.1 \times 10^5$ | pooled-barcode fitness                            | isolate-WGS          | <a href="https://doi.org/10.1038/s41467-026-73532-z">doi:10.1038/s41467-026-73532-z</a> |
| 15 | <b>Cooper 2010 (CE-MS amino-acid metabolome)</b>                                | Metabolite / precursor   | 4,700 YKO                    | 1                   | $4.7 \times 10^3$          | $9.4 \times 10^4$ | free amino-acid pools (capillary electrophoresis) | S288C-KO             | <a href="https://doi.org/10.1101/gr.105825.110">doi:10.1101/gr.105825.110</a>           |

Table 1, continued

| #  | Dataset                                           | Class                    | Genotypes                             | Env                              | Instances           | Meas.             | Phenotype                                                       | Sequence basis      | Link                                                                                        |
|----|---------------------------------------------------|--------------------------|---------------------------------------|----------------------------------|---------------------|-------------------|-----------------------------------------------------------------|---------------------|---------------------------------------------------------------------------------------------|
| 16 | Li 2016 (tRNA fitness landscape) •                | Deep mutational scan     | 65,000 tRNA variants                  | 1                                | $6.5 \times 10^4$ † | $6.5 \times 10^4$ | fitness                                                         | S288C+designed-edit | <a href="https://doi.org/10.1126/science.aae0568">doi:10.1126/science.aae0568</a>           |
| 17 | Aulakh 2025 (genome-scale ionome)                 | Metabolite / precursor   | 4,800 YKO                             | 1                                | $4.8 \times 10^3$   | $4.8 \times 10^4$ | intracellular metal-ion content (ICP-MS)                        | S288C-KO            | <a href="https://doi.org/10.1016/j.cels.2025.101319">doi:10.1016/j.cels.2025.101319</a>     |
| 18 | Peter 2018 (1,011 isolate genomes + phenome)      | Natural variation        | 1,011 isolates (971 phenotyped)       | 36 conditions                    | $3.5 \times 10^4$ † | $3.5 \times 10^4$ | growth fitness per condition                                    | isolate-WGS         | <a href="https://doi.org/10.1038/s41586-018-0030-5">doi:10.1038/s41586-018-0030-5</a>       |
| 19 | Domingo 2018 (tRNA double-mutant landscape) •     | Deep mutational scan     | 23,000 single + double mutants        | 1                                | $2.3 \times 10^4$ † | $2.3 \times 10^4$ | fitness                                                         | S288C+designed-edit | <a href="https://doi.org/10.1038/s41586-018-0170-7">doi:10.1038/s41586-018-0170-7</a>       |
| 20 | Zhang 2021 (YETI titratable overexpression)       | Metabolite / precursor   | 1,022 essential + 4,668 non-essential | beta-estradiol dose series       | $2.3 \times 10^4$ † | $2.3 \times 10^4$ | fitness across an induction gradient                            | S288C+tag           | <a href="https://doi.org/10.15252/msb.202010321">doi:10.15252/msb.202010321</a>             |
| 21 | Peeters 2021 (fermentation-trait QTL atlas)       | Natural variation        | 1,125 sequenced segregants            | 18 traits                        | $2.0 \times 10^4$ † | $2.0 \times 10^4$ | ethanol / glycerol / isobutanol titer, stress resistance, aroma | segregant-WGS       | PMID 34727964                                                                               |
| 22 | Van Leeuwen 2024 (short-chain organic acids)      | Tolerance / robustness   | 4,800 YKO                             | acetic, butyric, octanoic        | $1.4 \times 10^4$ † | $1.4 \times 10^4$ | fitness per acid                                                | S288C-KO            | PMC10903034                                                                                 |
| 23 | Chica 2026 (AutoDRY autophagy screen)             | Modality / backbone      | 4,760 KO + 1,159 DAmP                 | 2 (nitrogen starvation, replete) | $1.2 \times 10^4$ † | $1.2 \times 10^4$ | autophagy flux (deep-learning image classifier)                 | S288C-KO            | <a href="https://doi.org/10.1038/s41556-025-01837-0">doi:10.1038/s41556-025-01837-0</a>     |
| 24 | Sopko 2006 (genome-scale overexpression toxicity) | Metabolite / precursor   | 5,280 ORFs                            | GAL induced vs glucose           | $1.1 \times 10^4$ † | $1.1 \times 10^4$ | growth-rate reduction on induction                              | S288C+ORF-plasmid   | <a href="https://doi.org/10.1016/j.molcel.2005.12.011">doi:10.1016/j.molcel.2005.12.011</a> |
| 25 | Liu 2021 (tryptophan / isobutanol tolerance)      | Tolerance / robustness   | 5,006 YKO                             | +/- isobutanol                   | $1.0 \times 10^4$ † | $1.0 \times 10^4$ | colony-size fitness (ScreenMill imaging)                        | S288C-KO            | <a href="https://doi.org/10.1186/s13068-021-02048-z">doi:10.1186/s13068-021-02048-z</a>     |
| 26 | Kuroda 2019 (isobutanol-specific tolerance)       | Tolerance / robustness   | 4,800 YKO                             | isobutanol vs ethanol            | $9.6 \times 10^3$ † | $9.6 \times 10^3$ | isobutanol-specific fitness                                     | S288C-KO            | <a href="https://doi.org/10.1016/j.cels.2019.10.006">doi:10.1016/j.cels.2019.10.006</a>     |
| 27 | Puddu 2019 (WGS of the deletion collection)       | Modality / backbone      | 4,800 deletion strains                | 1                                | $4.8 \times 10^3$   | $4.8 \times 10^3$ | structural variants, CNV, aneuploidy                            | S288C-KO            | <a href="https://doi.org/10.1038/s41586-019-1549-9">doi:10.1038/s41586-019-1549-9</a>       |
| 28 | Turco 2023 (Yeast Phenome) <sup>L</sup>           | Tolerance / robustness   | 5,000                                 | 7,536 environments               | $3.8 \times 10^7$ † | $3.8 \times 10^7$ | growth / fitness per environment                                | S288C-KO            | <a href="https://doi.org/10.1126/sciadv.adg5702">doi:10.1126/sciadv.adg5702</a>             |
| 29 | N’Guessan 2025 (segregant scRNA-seq eQTL) •       | Expression / single cell | 4,500 profiled segregants             | 1                                | $4.5 \times 10^3$   | $2.7 \times 10^7$ | single-cell transcriptome + eQTL                                | segregant-WGS       | PMC12303567                                                                                 |
| 30 | Rossi 2021 (ChIP-exo protein architecture)        | Modality / backbone      | 400 tagged proteins                   | 1                                | 400                 | $2.4 \times 10^6$ | genome-wide binding location, near-bp resolution                | S288C+tag           | <a href="https://doi.org/10.1038/s41586-021-03314-8">doi:10.1038/s41586-021-03314-8</a>     |
| 31 | Piotrowski 2017 (MOSAIC diagnostic panel)         | Tolerance / robustness   | 157 diagnostic strains                | 13,524 compounds                 | $2.1 \times 10^6$ † | $2.1 \times 10^6$ | chemogenomic fitness score                                      | S288C-KO            | <a href="https://doi.org/10.1038/nchembio.2436">doi:10.1038/nchembio.2436</a>               |
| 32 | Hu 2007 (TF deletion expression compendium) •     | Expression / single cell | 269 TF deletion strains               | 1                                | 269                 | $1.6 \times 10^6$ | mRNA abundance                                                  | S288C-KO            | <a href="https://doi.org/10.1038/ng2012">doi:10.1038/ng2012</a>                             |

Table 1, continued

| #  | Dataset                                                      | Class                    | Genotypes                       | Env                              | Instances                  | Meas.             | Phenotype                                  | Sequence basis       | Link                             |
|----|--------------------------------------------------------------|--------------------------|---------------------------------|----------------------------------|----------------------------|-------------------|--------------------------------------------|----------------------|----------------------------------|
| 33 | Bloom 2025 (global epistasis in a yeast cross) •             | Natural variation        | large sequenced segregant panel | conditions                       | $1.0 \times 10^6 \ddagger$ | $1.0 \times 10^6$ | fitness                                    | segregant-WGS        | PMID 40679398                    |
| 34 | Gasch 2000 (environmental stress response)                   | Expression / single cell | wild type                       | 9 stress series, multi-timepoint | 150 $\ddagger$             | $9.0 \times 10^5$ | mRNA abundance time course                 | reference-only       | doi:10.1091/mbc.11.12.4241       |
| 35 | Decourty 2021 (RNA-metabolism genetic interactions)          | Modality / backbone      | 700,000 double mutants          | 1                                | $7.0 \times 10^5$          | $7.0 \times 10^5$ | genetic-interaction score                  | S288C-KO             | doi:10.1093/nar/gkab680          |
| 36 | Leutert 2023 (phosphoproteome x 101 conditions)              | Expression / single cell | wild type                       | 101 conditions                   | 101                        | $5.0 \times 10^5$ | phosphosite abundance                      | reference-only       | doi:10.1038/s41594-023-01115-3   |
| 37 | Cuperus 2017 (random 5' UTR library)                         | Regulatory DNA           | 500,000 random 5' UTRs          | 1                                | $5.0 \times 10^5 \ddagger$ | $5.0 \times 10^5$ | protein output (translation)               | S288C+reporter-locus | doi:10.1101/gr.224964.117        |
| 38 | Faure 2022 (doubledeepPCA) •                                 | Deep mutational scan     | 5e5 double mutants, 2 domains   | 1                                | $5.0 \times 10^5 \ddagger$ | $5.0 \times 10^5$ | binding and abundance                      | engineered-chassis   | doi:10.1038/s41586-022-04586-4   |
| 39 | Chong 2015 / CYCLOPs (single-cell proteome dynamics) •       | Modality / backbone      | 4,100 GFP-tagged ORFs           | cell-cycle / stress              | $2.5 \times 10^4 \ddagger$ | $4.2 \times 10^5$ | abundance + localization class             | S288C+tag            | doi:10.1016/j.cell.2015.04.051   |
| 40 | Brauer 2008 (growth-rate-controlled chemostat transcriptome) | Expression / single cell | wild type                       | 36 chemostats, 6 limitations     | 36                         | $2.2 \times 10^5$ | mRNA abundance                             | reference-only       | doi:10.1091/mbc.e07-08-0779      |
| 41 | Teyssonniere 2024 (species-wide trait survey)                | Natural variation        | 1,011 isolates                  | 200 traits                       | $2.0 \times 10^5 \ddagger$ | $2.0 \times 10^5$ | growth traits, expressivity and complexity | isolate-WGS          | doi:10.1371/journal.pgen.1011119 |
| 42 | Diss 2018 (Fos-Jun interaction double mutants) •             | Deep mutational scan     | 120,000 double mutants          | 1                                | $1.2 \times 10^5 \ddagger$ | $1.2 \times 10^5$ | protein-protein interaction strength       | engineered-chassis   | doi:10.7554/eLife.32472          |
| 43 | Warringer 2011 (trait variation across the SGRP panel)       | Natural variation        | 200 sequenced isolates          | 200 conditions                   | $4.0 \times 10^4 \ddagger$ | $1.2 \times 10^5$ | growth-curve parameters                    | isolate-WGS          | doi:10.1371/journal.pgen.1002111 |
| 44 | Dong 2021 (MAGIC + SAM biosensor)                            | CRISPR library screen    | 100,000 guides (a/i/d)          | 1                                | $1.0 \times 10^5 \ddagger$ | $1.0 \times 10^5$ | SAM biosensor fluorescence (FACS)          | S288C+guide          | doi:10.1016/j.ymben.2021.03.005  |
| 45 | Momen-Roknabadi 2020 (inducible CRISPRi library)             | CRISPR library screen    | genome-wide guide library       | nutrient limitations             | $9.0 \times 10^4 \ddagger$ | $9.0 \times 10^4$ | per-guide fitness under induction          | S288C+guide          | doi:10.1038/s42003-020-01452-9   |
| 46 | Keren 2013 (promoter activity across conditions)             | Regulatory DNA           | 900 native promoter reporters   | 10 conditions                    | $9.0 \times 10^3 \ddagger$ | $9.0 \times 10^4$ | promoter activity, time-resolved           | S288C+reporter-locus | doi:10.1038/msb.2013.59          |
| 47 | Hackett 2016 (SIMMER multi-omic flux)                        | Expression / single cell | wild type                       | 25 chemostat states              | 25                         | $7.5 \times 10^4$ | paired metabolome + proteome + flux        | reference-only       | doi:10.1126/science.aaf2786      |

Table 1, continued

| #  | Dataset                                                         | Class                  | Genotypes                              | Env                           | Instances                  | Meas.             | Phenotype                                          | Sequence basis      | Link                                                                                                    |
|----|-----------------------------------------------------------------|------------------------|----------------------------------------|-------------------------------|----------------------------|-------------------|----------------------------------------------------|---------------------|---------------------------------------------------------------------------------------------------------|
| 48 | McGlinchy 2021 (genome-scale CRISPRi library) •                 | CRISPR library screen  | 61,094 guides, 10/gene                 | continuous culture            | $6.1 \times 10^4$          | $6.1 \times 10^4$ | per-guide fitness                                  | S288C+guide         | <a href="https://doi.org/10.1186/s12864-021-07518-0">doi:10.1186/s12864-021-07518-0</a>                 |
| 49 | Bao 2018 (CHANGE single-nucleotide library)                     | CRISPR library screen  | tens of thousands of designed variants | inhibitor selection           | $6.0 \times 10^4 \ddagger$ | $6.0 \times 10^4$ | variant fitness (pooled barcode seq)               | S288C+designed-edit | <a href="https://doi.org/10.1038/nbt.4132">doi:10.1038/nbt.4132</a>                                     |
| 50 | Puchta 2016 (snoRNA U3 landscape) •                             | Deep mutational scan   | 60,000 U3 variants                     | 1                             | $6.0 \times 10^4 \ddagger$ | $6.0 \times 10^4$ | fitness                                            | S288C+designed-edit | <a href="https://doi.org/10.1126/science.aaf0965">doi:10.1126/science.aaf0965</a>                       |
| 51 | Yoshikawa 2011 (deletion + overexpression phenome)              | Tolerance / robustness | 4,800 KO + matched overexpression      | carbon sources / stress       | $5.8 \times 10^4 \ddagger$ | $5.8 \times 10^4$ | growth-rate fitness                                | S288C-KO            | <a href="https://doi.org/10.1534/g3.111.000695">doi:10.1534/g3.111.000695</a>                           |
| 52 | Ho 2009 (MoBY-ORF barcoded overexpression)                      | Metabolite / precursor | 5,100 barcoded ORFs                    | compound panel                | $5.1 \times 10^4 \ddagger$ | $5.1 \times 10^4$ | fitness per compound                               | S288C+ORF-plasmid   | <a href="https://doi.org/10.1038/nbt.1534">doi:10.1038/nbt.1534</a>                                     |
| 53 | Bloom 2013 (1,008 BYxRM segregants)                             | Natural variation      | 1,008 segregants, 11,623 markers       | 46 traits                     | $4.6 \times 10^4 \ddagger$ | $4.6 \times 10^4$ | quantitative growth traits                         | segregant-WGS       | <a href="https://doi.org/10.1038/nature11867">doi:10.1038/nature11867</a>                               |
| 54 | Zhu 2014 (kinase / phosphatase lipidomics)                      | Metabolite / precursor | 129 kinase + phosphatase KOs           | 1                             | 129                        | $2.6 \times 10^4$ | lipid class and species abundance                  | S288C-KO            | <a href="https://pubmed.ncbi.nlm.nih.gov/24196872/">PMC4196872</a>                                      |
| 55 | Xiao 2014 (genome-wide RNAi furfural tolerance)                 | Tolerance / robustness | genome-scale RNAi library              | furfural vs control           | $2.0 \times 10^4 \ddagger$ | $2.0 \times 10^4$ | knockdown enrichment under furfural                | S288C+guide         | <a href="https://doi.org/10.1186/1754-6834-7-78">doi:10.1186/1754-6834-7-78</a>                         |
| 56 | Crook 2016 (tunable RNAi, isobutanol + 1-butanol)               | Tolerance / robustness | genome-scale tunable RNAi              | isobutanol, 1-butanol         | $2.0 \times 10^4 \ddagger$ | $2.0 \times 10^4$ | dose-graded knockdown fitness under alcohol stress | S288C+guide         | <a href="https://pubmed.ncbi.nlm.nih.gov/27654654/">PMID 27654654</a>                                   |
| 57 | Douglas 2012 (barcoded overexpression fitness)                  | Metabolite / precursor | genome-scale barcoded ORFs             | baseline + stress             | $2.0 \times 10^4 \ddagger$ | $2.0 \times 10^4$ | pooled competitive fitness on induction            | S288C+ORF-plasmid   | <a href="https://doi.org/10.1534/g3.112.003400">doi:10.1534/g3.112.003400</a>                           |
| 58 | She 2018 (natural variant effects across a large isolate panel) | Natural variation      | 500 isolates                           | 40 conditions                 | $2.0 \times 10^4 \ddagger$ | $2.0 \times 10^4$ | growth traits                                      | isolate-WGS         | <a href="https://doi.org/10.1016/j.cell.2017.12.015">doi:10.1016/j.cell.2017.12.015</a>                 |
| 59 | Costello 2020 (bioreactor Bar-seq)                              | Tolerance / robustness | pooled YKO library                     | feed schemes / pH             | $1.9 \times 10^4 \ddagger$ | $1.9 \times 10^4$ | fitness under fed-batch cultivation                | S288C-KO            | <a href="https://doi.org/10.1186/s12934-020-01423-z">doi:10.1186/s12934-020-01423-z</a>                 |
| 60 | Roy 2018 (multiplexed precision editing)                        | CRISPR library screen  | thousands of designed edits            | 1                             | $1.6 \times 10^4 \ddagger$ | $1.6 \times 10^4$ | variant fitness                                    | S288C+designed-edit | <a href="https://doi.org/10.1038/nbt.4137">doi:10.1038/nbt.4137</a>                                     |
| 61 | Gerke 2017 (urea-cycle mQTL)                                    | Natural variation      | 147 diploid segregants                 | 1                             | 147                        | $1.5 \times 10^4$ | untargeted metabolite abundance                    | segregant-WGS       | <a href="https://doi.org/10.1534/genetics.117.201107">doi:10.1534/genetics.117.201107</a>               |
| 62 | Gurvitz (fatty-acid utilization screen)                         | Metabolite / precursor | 4,800 viable YKO                       | oleate, myristate, acetate    | $1.4 \times 10^4 \ddagger$ | $1.4 \times 10^4$ | fatty-acid utilization competence                  | S288C-KO            | <a href="https://embopress.org/doi/pdf/10.1038/msb4100051">embopress.org/doi/pdf/10.1038/msb4100051</a> |
| 63 | Mira 2010 (acetic acid tolerance, full collection)              | Tolerance / robustness | EUROSCARF collection                   | 70-110 mM acetic acid, pH 4.5 | $1.4 \times 10^4 \ddagger$ | $1.4 \times 10^4$ | acetic-acid tolerance                              | S288C-KO            | <a href="https://doi.org/10.1186/1475-2859-9-79">doi:10.1186/1475-2859-9-79</a>                         |
| 64 | Gameiro 2025 (AID2 degron libraries)                            | Modality / backbone    | >5,600 tagged ORFs                     | baseline + hydroxyurea        | $1.1 \times 10^4 \ddagger$ | $1.1 \times 10^4$ | degradation efficiency; fitness under stress       | S288C+tag           | <a href="https://doi.org/10.1083/jcb.202409007">doi:10.1083/jcb.202409007</a>                           |



Table 1, continued

| #  | Dataset                                                 | Class                  | Genotypes                               | Env                    | Instances                  | Meas.             | Phenotype                                              | Sequence basis       | Link                                                                                        |
|----|---------------------------------------------------------|------------------------|-----------------------------------------|------------------------|----------------------------|-------------------|--------------------------------------------------------|----------------------|---------------------------------------------------------------------------------------------|
| 65 | Si 2017 (RAGE RNAi, isobutanol titer)                   | CRISPR library screen  | genome-scale RNAi, iterative rounds     | isobutanol selection   | $1.0 \times 10^4 \ddagger$ | $1.0 \times 10^4$ | construct enrichment; isobutanol titer for top strains | S288C+guide          | <a href="https://doi.org/10.1038/ncomms15187">doi:10.1038/ncomms15187</a>                   |
| 66 | Hamedirad 2018 (RAGE xylose utilization)                | CRISPR library screen  | genome-scale RNAi                       | xylose-limited         | $1.0 \times 10^4 \ddagger$ | $1.0 \times 10^4$ | enrichment; xylose consumption rate                    | S288C+guide          | <a href="https://doi.org/10.1002/bit.26570">doi:10.1002/bit.26570</a>                       |
| 67 | Newman 2006 (protein noise, GFP library) •              | Modality / backbone    | 2,500 GFP-tagged strains                | rich / minimal         | $5.0 \times 10^3 \dagger$  | $1.0 \times 10^4$ | protein abundance and cell-to-cell noise               | S288C+tag            | <a href="https://doi.org/10.1038/nature04785">doi:10.1038/nature04785</a>                   |
| 68 | Pereira 2014 (wheat-straw hydrolysate)                  | Tolerance / robustness | EUROSCARF deletion collection           | hydrolysate vs control | $9.6 \times 10^3 \dagger$  | $9.6 \times 10^3$ | tolerance to industrial hydrolysate                    | S288C-KO             | <a href="https://doi.org/10.1007/s10295-014-1519-z">doi:10.1007/s10295-014-1519-z</a>       |
| 69 | Endo 2008 (vanillin tolerance)                          | Tolerance / robustness | diploid deletion collection             | vanillin vs control    | $9.6 \times 10^3 \dagger$  | $9.6 \times 10^3$ | vanillin sensitivity                                   | S288C-KO             | <a href="https://doi.org/10.1128/AEM.01541-08">doi:10.1128/AEM.01541-08</a>                 |
| 70 | Sharifpoor 2012 (kinome synthetic dosage lethality)     | Modality / backbone    | 92 kinase queries, genome-wide array    | 1                      | $8.7 \times 10^3$          | $8.7 \times 10^3$ | genetic interaction / SDL score                        | S288C-KO             | <a href="https://doi.org/10.1101/gr.129213.111">doi:10.1101/gr.129213.111</a>               |
| 71 | Ambroset 2014 (metabolite QTL)                          | Natural variation      | 100 segregants                          | 1                      | 100                        | $7.4 \times 10^3$ | 74 metabolite concentrations (LC-MS/MS)                | segregant-WGS        | <a href="https://doi.org/10.1371/journal.pgen.1004142">doi:10.1371/journal.pgen.1004142</a> |
| 72 | Mavor 2016 (ubiquitin DMS across conditions)            | Deep mutational scan   | 1,400 ubiquitin variants                | chemical stresses      | $7.0 \times 10^3 \dagger$  | $7.0 \times 10^3$ | fitness                                                | S288C+designed-edit  | <a href="https://doi.org/10.7554/eLife.15802">doi:10.7554/eLife.15802</a>                   |
| 73 | Blank 2005 (13C metabolic flux)                         | Metabolite / precursor | 200 viable null mutants                 | glucose minimal        | 200 ‡                      | $6.0 \times 10^3$ | intracellular flux distribution (13C-MFA)              | S288C-KO             | <a href="https://doi.org/10.1186/gb-2005-6-6-r49">doi:10.1186/gb-2005-6-6-r49</a>           |
| 74 | Flynn 2020 (Hsp90 DMS across environments)              | Deep mutational scan   | 1,000 Hsp90 variants                    | stress conditions      | $6.0 \times 10^3 \dagger$  | $6.0 \times 10^3$ | fitness                                                | S288C+designed-edit  | <a href="https://doi.org/10.7554/eLife.53810">doi:10.7554/eLife.53810</a>                   |
| 75 | Ho 2018 (unified absolute protein abundance)            | Modality / backbone    | wild type                               | 1                      | $5.9 \times 10^3$          | $5.9 \times 10^3$ | absolute abundance, molecules/cell                     | reference-only       | <a href="https://doi.org/10.1016/j.cels.2017.12.004">doi:10.1016/j.cels.2017.12.004</a>     |
| 76 | Renganaath 2020 (natural promoter-variant MPRA) •       | Regulatory DNA         | 5,832 natural variants, 2,503 promoters | 1                      | $5.8 \times 10^3$          | $5.8 \times 10^3$ | allele-specific promoter activity                      | S288C+reporter-locus | <a href="https://doi.org/10.7554/eLife.62669">doi:10.7554/eLife.62669</a>                   |
| 77 | Chang 2012 (organic-acid production screen)             | Metabolite / precursor | 4,800 YKO                               | 1                      | $4.8 \times 10^3$          | $4.8 \times 10^3$ | acetate / pyruvate / succinate halo (pH indicator)     | S288C-KO             | <a href="https://pubmed.ncbi.nlm.nih.gov/22277779/">PMID 22277779</a>                       |
| 78 | Trikka 2015 (carotenogenic heterozygous screen)         | Metabolite / precursor | 4,700 heterozygous deletions            | 1                      | $4.7 \times 10^3$          | $4.7 \times 10^3$ | carotenoid color; sclareol titer for top strains       | S288C-KO/het         | <a href="https://doi.org/10.1186/s12934-015-0246-0">doi:10.1186/s12934-015-0246-0</a>       |
| 79 | Mukherjee 2021 (CRISPRi essential genes x acetic acid)  | Tolerance / robustness | 1,100 essential genes                   | acetic acid vs control | $2.2 \times 10^3 \dagger$  | $2.2 \times 10^3$ | fitness under acid stress                              | S288C+guide          | <a href="https://doi.org/10.1128/mSystems.00410-21">doi:10.1128/mSystems.00410-21</a>       |
| 80 | Sardi 2018 (natural variation in hydrolysate tolerance) | Tolerance / robustness | sequenced isolate / segregant panel     | hydrolysate toxins     | $2.0 \times 10^3 \ddagger$ | $2.0 \times 10^3$ | tolerance                                              | isolate-WGS          | <a href="https://doi.org/10.1371/journal.pgen.1007217">doi:10.1371/journal.pgen.1007217</a> |



Table 1, continued

| #  | Dataset                                                         | Class                    | Genotypes                             | Env                           | Instances           | Meas.             | Phenotype                                  | Sequence basis      | Link                                                                                        |
|----|-----------------------------------------------------------------|--------------------------|---------------------------------------|-------------------------------|---------------------|-------------------|--------------------------------------------|---------------------|---------------------------------------------------------------------------------------------|
| 81 | <b>Breslow 2008 (DAmP hypomorphic collection)</b>               | Modality / backbone      | 842 haploid + 970 diploid essential   | 1                             | $1.8 \times 10^3$   | $1.8 \times 10^3$ | competitive fitness                        | S288C+tag           | <a href="https://doi.org/10.1038/nmeth.1234">doi:10.1038/nmeth.1234</a>                     |
| 82 | <b>Kofoed 2015 (barcoded ts alleles)</b>                        | Modality / backbone      | 600+ barcoded ts alleles              | permissive / restrictive      | $1.2 \times 10^3$ † | $1.2 \times 10^3$ | ts severity (growth ratio)                 | S288C+designed-edit | <a href="https://doi.org/10.1534/g3.115.019174">doi:10.1534/g3.115.019174</a>               |
| 83 | <b>Lian 2017 (CRISPR-AID, beta-carotene)</b>                    | CRISPR library screen    | combinatorial a/i/d triples           | 1                             | 200‡                | 200               | beta-carotene titer                        | engineered-chassis  | <a href="https://doi.org/10.1038/s41467-017-01695-x">doi:10.1038/s41467-017-01695-x</a>     |
| 84 | <b>Zhang 2022 (GCE-SCRaMbLE recombination outcomes) •</b>       | Combinatorial genome     | 200 sequenced SCRaMbLE genomes        | 1                             | 200‡                | 200               | rearrangement structure + fitness          | isolate-WGS         | <a href="https://doi.org/10.1038/s41467-022-33606-0">doi:10.1038/s41467-022-33606-0</a>     |
| 85 | <b>Cubillos 2017 (nitrogen-consumption QTL)</b>                 | Natural variation        | 165 sequenced F12 segregants          | nitrogen-limited fermentation | 165                 | 165               | nitrogen consumption                       | segregant-WGS       | <a href="https://doi.org/10.1534/g3.117.042127">doi:10.1534/g3.117.042127</a>               |
| 86 | <b>Hillenmeyer 2010 (systematic fitness reanalysis)</b>         | Tolerance / robustness   | het + hom collections                 | 400 conditions                | $2.4 \times 10^6$ ‡ | $2.4 \times 10^6$ | fitness-defect score                       | S288C-KO            | <a href="https://doi.org/10.1186/gb-2010-11-3-r30">doi:10.1186/gb-2010-11-3-r30</a>         |
| 87 | <b>Smith 2008 (gene-environment interaction eQTL) •</b>         | Natural variation        | 109 segregants                        | glucose / ethanol             | 218†                | $1.3 \times 10^6$ | mRNA abundance                             | segregant-WGS       | <a href="https://doi.org/10.1371/journal.pbio.0060083">doi:10.1371/journal.pbio.0060083</a> |
| 88 | <b>Lenstra 2011 (chromatin regulator deletion expression) •</b> | Expression / single cell | 165 chromatin-regulator deletions     | 1                             | 165                 | $9.9 \times 10^5$ | mRNA abundance                             | S288C-KO            | <a href="https://doi.org/10.1016/j.molcel.2011.03.026">doi:10.1016/j.molcel.2011.03.026</a> |
| 89 | <b>Braberg 2020 (point-mutant E-MAP)</b>                        | Modality / backbone      | 350 point mutants x genome-wide array | 1                             | $5.0 \times 10^5$   | $5.0 \times 10^5$ | residue-level genetic interaction          | S288C+designed-edit | <a href="https://doi.org/10.1126/science.aaz4910">doi:10.1126/science.aaz4910</a>           |
| 90 | <b>Breker 2013 (GFP library under stress) •</b>                 | Modality / backbone      | 4,159 GFP-tagged strains              | stress conditions             | $1.2 \times 10^4$ † | $2.1 \times 10^5$ | abundance + localization, per cell         | S288C+tag           | <a href="https://doi.org/10.1083/jcb.201301120">doi:10.1083/jcb.201301120</a>               |
| 91 | <b>Tkach 2012 (protein relocation under DNA damage) •</b>       | Modality / backbone      | GFP collection                        | DNA-damaging agents           | $1.2 \times 10^4$ † | $2.0 \times 10^5$ | localization and abundance change          | S288C+tag           | <a href="https://doi.org/10.1038/ncb2549">doi:10.1038/ncb2549</a>                           |
| 92 | <b>Ehrenreich 2010 (X-QTL bulk segregant mapping)</b>           | Natural variation        | 1e6 pooled segregants                 | 17 selective conditions       | 17                  | $1.9 \times 10^5$ | allele-frequency shift (pooled)            | segregant-WGS       | <a href="https://doi.org/10.1038/nature08923">doi:10.1038/nature08923</a>                   |
| 93 | <b>Skelly 2013 (expression variation across isolates)</b>       | Natural variation        | 22 sequenced isolates                 | 1                             | 22                  | $1.8 \times 10^5$ | transcriptome + proteome                   | isolate-WGS         | <a href="https://doi.org/10.1101/gr.155762.113">doi:10.1101/gr.155762.113</a>               |
| 94 | <b>Denervaud 2013 (microfluidic GFP dynamics) •</b>             | Modality / backbone      | 1,000 GFP strains                     | dynamic environments          | $4.0 \times 10^3$ † | $6.8 \times 10^4$ | time-resolved single-cell abundance        | S288C+tag           | <a href="https://doi.org/10.1073/pnas.1308265110">doi:10.1073/pnas.1308265110</a>           |
| 95 | <b>Brogaard 2012 (chemical nucleosome map)</b>                  | Modality / backbone      | wild type                             | 1                             | 1                   | $6.7 \times 10^4$ | nucleosome center positions, bp resolution | reference-only      | <a href="https://doi.org/10.1038/nature11142">doi:10.1038/nature11142</a>                   |

Table 1, continued

| #   | Dataset                                                           | Class                    | Genotypes                          | Env                       | Instances           | Meas.             | Phenotype                                     | Sequence basis      | Link                                                                                        |
|-----|-------------------------------------------------------------------|--------------------------|------------------------------------|---------------------------|---------------------|-------------------|-----------------------------------------------|---------------------|---------------------------------------------------------------------------------------------|
| 96  | <b>Xia 2022 (proteome allocation across conditions)</b>           | Modality / backbone      | wild type                          | chemostat conditions      | 30‡                 | $6.0 \times 10^4$ | absolute proteome, resource allocation        | reference-only      | <a href="https://doi.org/10.1038/s41467-022-30513-2">doi:10.1038/s41467-022-30513-2</a>     |
| 97  | <b>Melamed 2013 (PAB1 RRM domain scan)</b>                        | Deep mutational scan     | 40,000 PAB1 variants               | 1                         | $4.0 \times 10^4$ ‡ | $4.0 \times 10^4$ | fitness                                       | S288C+designed-edit | <a href="https://doi.org/10.1261/rna.040709.113">doi:10.1261/rna.040709.113</a>             |
| 98  | <b>De Chiara 2022 (domestication and genome instability)</b>      | Natural variation        | 1,011-isolate panel, reanalyzed    | conditions                | $3.6 \times 10^4$ ‡ | $3.6 \times 10^4$ | growth, aneuploidy, instability               | isolate-WGS         | <a href="https://doi.org/10.1038/s41559-022-01671-9">doi:10.1038/s41559-022-01671-9</a>     |
| 99  | <b>Guo 2018 (CRISPR-Cas9 tiling of essential genes) •</b>         | CRISPR library screen    | 3e4 tiling guides                  | 1                         | $3.0 \times 10^4$ ‡ | $3.0 \times 10^4$ | per-guide fitness                             | S288C+guide         | <a href="https://doi.org/10.1038/nbt.4147">doi:10.1038/nbt.4147</a>                         |
| 100 | <b>Yvert 2013 (single-cell trait variation across isolates) •</b> | Natural variation        | 37 isolates                        | 1                         | 37                  | $1.9 \times 10^4$ | single-cell morphology (CalMorph)             | isolate-WGS         | <a href="https://doi.org/10.1186/1752-0509-7-54">doi:10.1186/1752-0509-7-54</a>             |
| 101 | <b>Lahtvee 2017 (absolute proteome across conditions)</b>         | Modality / backbone      | wild type                          | 9 growth conditions       | 9                   | $1.8 \times 10^4$ | absolute protein abundance + turnover         | reference-only      | <a href="https://doi.org/10.1016/j.cels.2017.03.003">doi:10.1016/j.cels.2017.03.003</a>     |
| 102 | <b>Weill 2018 (SWAT libraries)</b>                                | Modality / backbone      | 5,500 acceptor-tagged strains      | 1                         | $5.5 \times 10^3$   | $1.6 \times 10^4$ | abundance, localization, topology             | S288C+tag           | <a href="https://doi.org/10.1038/s41592-018-0044-9">doi:10.1038/s41592-018-0044-9</a>       |
| 103 | <b>Kinsler 2020 (barcoded pleiotropy across environments)</b>     | Natural variation        | 300 sequenced adaptive clones      | 45 environments           | $1.4 \times 10^4$ † | $1.4 \times 10^4$ | fitness                                       | isolate-WGS         | <a href="https://doi.org/10.7554/eLife.61271">doi:10.7554/eLife.61271</a>                   |
| 104 | <b>Nadal-Ribelles 2019 (sensitive yeast scRNA-seq) •</b>          | Expression / single cell | clonal wild type                   | baseline + stress         | 2‡                  | $1.2 \times 10^4$ | single-cell transcriptome                     | reference-only      | <a href="https://doi.org/10.1038/s41564-018-0346-9">doi:10.1038/s41564-018-0346-9</a>       |
| 105 | <b>Gasch 2017 (single-cell stress heterogeneity) •</b>            | Expression / single cell | wild type                          | stress vs unstressed      | 2‡                  | $1.2 \times 10^4$ | single-cell transcriptome                     | reference-only      | <a href="https://doi.org/10.1371/journal.pbio.2004050">doi:10.1371/journal.pbio.2004050</a> |
| 106 | <b>Pelechano 2013 (transcript isoform landscape)</b>              | Modality / backbone      | wild type                          | glucose / galactose       | 2                   | $1.2 \times 10^4$ | transcript isoform boundaries and abundance   | reference-only      | <a href="https://doi.org/10.1038/nature12121">doi:10.1038/nature12121</a>                   |
| 107 | <b>Deutschbauer 2005 (haploinsufficiency, genome-wide)</b>        | Tolerance / robustness   | heterozygous diploid collection    | rich / minimal            | $1.2 \times 10^4$ † | $1.2 \times 10^4$ | haploinsufficiency fitness                    | S288C-KO/het        | <a href="https://doi.org/10.1534/genetics.104.036871">doi:10.1534/genetics.104.036871</a>   |
| 108 | <b>Valenti 2025 (SWAT degon + GFP collection)</b>                 | Modality / backbone      | proteome-wide SWAT library         | depleted / not            | $1.0 \times 10^4$ ‡ | $1.0 \times 10^4$ | depletion efficiency + abundance/localization | S288C+tag           | <a href="https://doi.org/10.1083/jcb.202409050">doi:10.1083/jcb.202409050</a>               |
| 109 | <b>Sousa 2013 (acetate resistance overexpression screen)</b>      | Tolerance / robustness   | genome-wide overexpression library | sodium acetate vs control | $1.0 \times 10^4$ † | $1.0 \times 10^4$ | acetate resistance                            | S288C+ORF-plasmid   | PMID 23262128                                                                               |
| 110 | <b>Jaffe 2019 (multiplexed CRISPR interference epistasis) •</b>   | CRISPR library screen    | 1e4 guide pairs                    | 1                         | $1.0 \times 10^4$ ‡ | $1.0 \times 10^4$ | fitness / genetic interaction                 | S288C+guide         | <a href="https://doi.org/10.1534/g3.119.400323">doi:10.1534/g3.119.400323</a>               |

Table 1, continued

| #   | Dataset                                                                 | Class                  | Genotypes                           | Env                                 | Instances                  | Meas.             | Phenotype                                  | Sequence basis       | Link                                                                                        |
|-----|-------------------------------------------------------------------------|------------------------|-------------------------------------|-------------------------------------|----------------------------|-------------------|--------------------------------------------|----------------------|---------------------------------------------------------------------------------------------|
| 111 | <b>Sadhu 2016</b><br>(CRISPR-directed causal variant mapping)           | Natural variation      | 1,000 targeted recombinants         | conditions                          | $1.0 \times 10^4 \ddagger$ | $1.0 \times 10^4$ | fitness                                    | segregant-WGS        | <a href="https://doi.org/10.1126/science.aaf5124">doi:10.1126/science.aaf5124</a>           |
| 112 | <b>Weinberg 2016</b> (ribosome profiling, improved)                     | Modality / backbone    | wild type                           | 1                                   | 1                          | $6.0 \times 10^3$ | ribosome occupancy, translation efficiency | reference-only       | <a href="https://doi.org/10.1016/j.celrep.2016.01.043">doi:10.1016/j.celrep.2016.01.043</a> |
| 113 | <b>Ghaemmaghami 2003</b> (absolute protein abundance)                   | Modality / backbone    | 4,251 TAP-tagged ORFs               | 1                                   | $4.3 \times 10^3$          | $4.3 \times 10^3$ | absolute abundance (molecules/cell)        | S288C+tag            | <a href="https://doi.org/10.1038/nature02046">doi:10.1038/nature02046</a>                   |
| 114 | <b>Huh 2003</b> (GFP localization atlas)                                | Modality / backbone    | 4,156 GFP-tagged ORFs               | 1                                   | $4.2 \times 10^3$          | $4.2 \times 10^3$ | subcellular localization (22 classes)      | S288C+tag            | <a href="https://doi.org/10.1038/nature02026">doi:10.1038/nature02026</a>                   |
| 115 | <b>Michaelis 2023</b> (protein interactome)                             | Modality / backbone    | proteome-scale AP-MS baits          | 1                                   | $4.0 \times 10^3 \ddagger$ | $4.0 \times 10^3$ | protein-protein interaction + interface    | S288C+tag            | <a href="https://doi.org/10.1038/s41586-023-06739-5">doi:10.1038/s41586-023-06739-5</a>     |
| 116 | <b>Christiano 2014</b> (protein turnover)                               | Modality / backbone    | wild type                           | 1                                   | $3.9 \times 10^3 \ddagger$ | $3.9 \times 10^3$ | protein half-life                          | reference-only       | <a href="https://doi.org/10.1016/j.celrep.2014.10.065">doi:10.1016/j.celrep.2014.10.065</a> |
| 117 | <b>Boer 2010</b> (metabolome across nutrient limitations)               | Metabolite / precursor | wild type                           | nutrient limitations x growth rates | 25                         | $2.5 \times 10^3$ | intracellular metabolite concentrations    | reference-only       | <a href="https://doi.org/10.1091/mbc.e09-07-0597">doi:10.1091/mbc.e09-07-0597</a>           |
| 118 | <b>Kitzman 2015</b> (GAL4 DMS)                                          | Deep mutational scan   | 1,200 GAL4 variants                 | selective / permissive              | $2.4 \times 10^3 \dagger$  | $2.4 \times 10^3$ | transcription-factor activity              | S288C+designed-edit  | <a href="https://doi.org/10.1038/nmeth.3223">doi:10.1038/nmeth.3223</a>                     |
| 119 | <b>Sharon 2012</b> (designed promoter library)                          | Regulatory DNA         | 2,000 designed promoters            | 1                                   | $2.0 \times 10^3 \ddagger$ | $2.0 \times 10^3$ | expression (fluorescence)                  | S288C+reporter-locus | <a href="https://doi.org/10.1038/nbt.2205">doi:10.1038/nbt.2205</a>                         |
| 120 | <b>Tengolics 2024</b> (domestication metabolome)                        | Natural variation      | 17 <i>S. cerevisiae</i> populations | 1                                   | 17                         | $1.6 \times 10^3$ | 19 amino acids + 78 other metabolites      | isolate-WGS          | <a href="https://doi.org/10.1073/pnas.2313354121">doi:10.1073/pnas.2313354121</a>           |
| 121 | <b>Sadhu 2018</b> (CRISPR-directed mitotic recombination)               | CRISPR library screen  | 1,500 recombinant genotypes         | 1                                   | $1.5 \times 10^3 \ddagger$ | $1.5 \times 10^3$ | fitness / trait                            | segregant-WGS        | <a href="https://doi.org/10.7554/eLife.41090">doi:10.7554/eLife.41090</a>                   |
| 122 | <b>Duveau 2021</b> (TDH3 promoter allele series)                        | Regulatory DNA         | 250 promoter alleles                | carbon sources                      | $1.0 \times 10^3 \ddagger$ | $1.0 \times 10^3$ | expression level, noise, fitness           | S288C+designed-edit  | <a href="https://doi.org/10.7554/eLife.67806">doi:10.7554/eLife.67806</a>                   |
| 123 | <b>Strope 2015</b> (100 clinical and natural genomes)                   | Natural variation      | 100 sequenced isolates              | phenotype panel                     | $1.0 \times 10^3 \ddagger$ | $1.0 \times 10^3$ | growth and clinical-relevant traits        | isolate-WGS          | <a href="https://doi.org/10.1101/gr.185538.114">doi:10.1101/gr.185538.114</a>               |
| 124 | <b>Schulte 2023</b> (mitochondrial complexome)                          | Modality / backbone    | 900 mitochondrial proteins          | 1                                   | 900                        | 900               | complex assembly profile                   | reference-only       | <a href="https://doi.org/10.1038/s41586-022-05641-w">doi:10.1038/s41586-022-05641-w</a>     |
| 125 | <b>Metzger 2015</b> (promoter mutation effects on expression and noise) | Regulatory DNA         | 235 promoter mutants                | 1                                   | 235 $\ddagger$             | 470               | expression mean and noise                  | S288C+designed-edit  | <a href="https://doi.org/10.1038/nature14424">doi:10.1038/nature14424</a>                   |
| 126 | <b>Venkataram 2016</b> (barcoded adaptive lineages, sequenced)          | Natural variation      | 400 sequenced adaptive clones       | glucose-limited                     | 400 $\ddagger$             | 400               | fitness                                    | isolate-WGS          | <a href="https://doi.org/10.1016/j.cell.2016.08.002">doi:10.1016/j.cell.2016.08.002</a>     |

Table 1, continued

| #   | Dataset                                                                                | Class                    | Genotypes                                     | Env                       | Instances           | Meas.             | Phenotype                                     | Sequence basis     | Link                                                                                        |
|-----|----------------------------------------------------------------------------------------|--------------------------|-----------------------------------------------|---------------------------|---------------------|-------------------|-----------------------------------------------|--------------------|---------------------------------------------------------------------------------------------|
| 127 | <b>Jakociunas 2021 (degron-tuned terpene flux)</b>                                     | Metabolite / precursor   | 30 degron strains                             | induction levels          | 90‡                 | 270               | farnesol / geraniol / nerolidol titer         | engineered-chassis | <a href="https://doi.org/10.1002/bit.27735">doi:10.1002/bit.27735</a>                       |
| 128 | <b>Shen 2016 (SCRaMbLE genome minimization)</b>                                        | Combinatorial genome     | 100 SCRaMbLEd genomes                         | selective / permissive    | 200‡                | 200               | viability and fitness                         | isolate-WGS        | <a href="https://doi.org/10.1101/gr.193433.115">doi:10.1101/gr.193433.115</a>               |
| 129 | <b>Blount 2018 (SCRaMbLE for phenotype improvement)</b>                                | Combinatorial genome     | 100 rearranged genomes                        | selection conditions      | 200‡                | 200               | growth and product phenotype                  | isolate-WGS        | <a href="https://doi.org/10.1038/s41467-018-03143-w">doi:10.1038/s41467-018-03143-w</a>     |
| 130 | <b>Eder 2020 (flux QTL)</b>                                                            | Natural variation        | segregant panel (n unconfirmed)               | wine fermentation         | 100‡                | 100               | modeled intracellular flux                    | segregant-WGS      | <a href="https://pubmed.ncbi.nlm.nih.gov/32034164/">PMID 32034164</a>                       |
| 131 | <b>Liti 2009 (population genomics of <i>S. cerevisiae</i> and <i>S. paradoxus</i>)</b> | Natural variation        | 70 sequenced isolates                         | 1                         | 70‡                 | 70                | genome sequence and population structure      | isolate-WGS        | <a href="https://doi.org/10.1038/nature07743">doi:10.1038/nature07743</a>                   |
| 132 | <b>Yue 2017 (wild vs domesticated assemblies)</b>                                      | Natural variation        | 12 long-read assemblies                       | n/a                       | 12                  | 12                | genome structure, TE and subtelomere dynamics | isolate-WGS        | <a href="https://doi.org/10.1038/ng.3847">doi:10.1038/ng.3847</a>                           |
| 133 | <b>Brettner 2024 (ultra-high-throughput yeast scRNA-seq) •</b>                         | Expression / single cell | method                                        | method                    | –‡                  | –                 | single-cell transcriptome                     | reference-only     | <a href="https://doi.org/10.1002/yea.3927">doi:10.1002/yea.3927</a>                         |
| 134 | <b>Mulleder 2012 (prototrophic deletion collection)</b>                                | Modality / backbone      | 5,185 prototrophic + 839 titratable essential | 1                         | –‡                  | –                 | strain resource (no phenotype)                | S288C-KO           | <a href="https://doi.org/10.1038/nbt.2442">doi:10.1038/nbt.2442</a>                         |
| 135 | <b>Zackrisson 2016 (Scan-o-matic growth curves)</b>                                    | Modality / backbone      | YKO-compatible platform                       | panel-dependent           | –‡                  | –                 | full growth curve (rate, lag, yield)          | S288C-KO           | <a href="https://doi.org/10.1534/g3.116.032342">doi:10.1534/g3.116.032342</a>               |
| 136 | <b>Spellman 1998 (cell-cycle transcriptome)</b>                                        | Expression / single cell | wild type, synchronized                       | 4 synchronization methods | 77                  | $4.6 \times 10^5$ | mRNA abundance time course                    | reference-only     | <a href="https://doi.org/10.1091/mbc.9.12.3273">doi:10.1091/mbc.9.12.3273</a>               |
| 137 | <b>Levy 2008 (morphology and phenotypic capacitance) •</b>                             | Modality / backbone      | 1,700 deletion strains                        | 1                         | $1.7 \times 10^3$ ‡ | $3.4 \times 10^5$ | single-cell morphology variability            | S288C-KO           | <a href="https://doi.org/10.1371/journal.pbio.0060264">doi:10.1371/journal.pbio.0060264</a> |
| 138 | <b>Causton 2001 (environmental change transcriptome)</b>                               | Expression / single cell | wild type                                     | environmental changes     | 40‡                 | $2.4 \times 10^5$ | mRNA abundance                                | reference-only     | <a href="https://doi.org/10.1091/mbc.12.2.323">doi:10.1091/mbc.12.2.323</a>                 |
| 139 | <b>Gasch 2001 (DNA damage response transcriptome)</b>                                  | Expression / single cell | wild type + checkpoint mutants                | damaging agents           | 24‡                 | $1.4 \times 10^5$ | mRNA abundance                                | S288C-KO           | <a href="https://doi.org/10.1091/mbc.12.10.2987">doi:10.1091/mbc.12.10.2987</a>             |
| 140 | <b>Handfield 2013 (image-based localization change)</b>                                | Modality / backbone      | GFP collection                                | baseline / perturbed      | $8.0 \times 10^3$ ‡ | $1.4 \times 10^5$ | localization change score                     | S288C+tag          | <a href="https://doi.org/10.1371/journal.pcbi.1003085">doi:10.1371/journal.pcbi.1003085</a> |
| 141 | <b>Kaplan 2009 (nucleosome sequence preferences)</b>                                   | Modality / backbone      | wild type (in vitro + in vivo)                | in vitro / in vivo        | 2                   | $1.3 \times 10^5$ | nucleosome occupancy                          | reference-only     | <a href="https://doi.org/10.1038/nature07667">doi:10.1038/nature07667</a>                   |

Table 1, continued

| #                                                                                   | Dataset                                                     | Class                  | Genotypes                         | Env                              | Instances           | Meas.             | Phenotype                               | Sequence basis     | Link                                                                                                                     |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------|------------------------|-----------------------------------|----------------------------------|---------------------|-------------------|-----------------------------------------|--------------------|--------------------------------------------------------------------------------------------------------------------------|
| 142                                                                                 | Giaever 2004 (chemogenomic drug-target profiling)           | Tolerance / robustness | het + hom collections             | 10 compounds                     | $5.9 \times 10^4$ ‡ | $5.9 \times 10^4$ | fitness                                 | S288C-KO           | <a href="https://doi.org/10.1073/pnas.0307490100">doi:10.1073/pnas.0307490100</a>                                        |
| 143                                                                                 | Yu 2021 (proteome and metabolome under nitrogen limitation) | Metabolite / precursor | wild type                         | nitrogen-limited chemostats      | 10‡                 | $2.5 \times 10^4$ | proteome + metabolome                   | reference-only     | <a href="https://doi.org/10.7554/eLife.65722">doi:10.7554/eLife.65722</a>                                                |
| 144                                                                                 | Paulo 2016 (TMT proteome across carbon sources)             | Modality / backbone    | wild type                         | carbon sources                   | 9                   | $2.2 \times 10^4$ | protein abundance                       | reference-only     | <a href="https://doi.org/10.1091/mbc.E16-03-0184">doi:10.1091/mbc.E16-03-0184</a>                                        |
| 145                                                                                 | Gibney 2013 (osmotic and glucose stress fitness)            | Tolerance / robustness | YKO collection                    | osmotic / glucose                | $1.9 \times 10^4$ ‡ | $1.9 \times 10^4$ | fitness                                 | S288C-KO           | <a href="https://doi.org/10.1073/pnas.1318100110">doi:10.1073/pnas.1318100110</a>                                        |
| 146                                                                                 | Meurer 2018 (genome-wide GFP library, rebuilt)              | Modality / backbone    | 4,000 tagged strains              | 1                                | $4.0 \times 10^3$ ‡ | $1.2 \times 10^4$ | abundance and localization              | S288C+tag          | <a href="https://doi.org/10.1038/s41592-018-0044-9">doi:10.1038/s41592-018-0044-9</a>                                    |
| 147                                                                                 | Bennett 2001 (ionizing-radiation resistance)                | Tolerance / robustness | 4,800 YKO                         | irradiated vs control            | $9.6 \times 10^3$ ‡ | $9.6 \times 10^3$ | radiation sensitivity                   | S288C-KO           | <a href="https://doi.org/10.1038/ng778">doi:10.1038/ng778</a>                                                            |
| 148                                                                                 | Sugiyama (lactic-acid tolerance)                            | Tolerance / robustness | 4,800 YKO                         | 4 percent lactic acid vs control | $9.6 \times 10^3$ ‡ | $9.6 \times 10^3$ | acid tolerance                          | S288C-KO           | <a href="http://nistr.or.jp/wp-content/uploads/NISR06sugiyama.pdf">nistr.or.jp/wp-content/uploads/NISR06sugiyama.pdf</a> |
| 149                                                                                 | Sousa 2013 (acetic-acid programmed cell death screen)       | Tolerance / robustness | YKO haploid collection            | acetic acid vs control           | $9.6 \times 10^3$ ‡ | $9.6 \times 10^3$ | death / survival (regulated cell death) | S288C-KO           | PMID 24286259                                                                                                            |
| 150                                                                                 | Nagalakshmi 2008 (RNA-seq transcriptome annotation)         | Modality / backbone    | wild type                         | 1                                | 1                   | $6.0 \times 10^3$ | transcript structure and abundance      | reference-only     | <a href="https://doi.org/10.1126/science.1158441">doi:10.1126/science.1158441</a>                                        |
| 151                                                                                 | Yofe 2016 (SWAT N-terminal library)                         | Modality / backbone    | 1,800 N-terminally tagged strains | 1                                | $1.8 \times 10^3$ ‡ | $5.4 \times 10^3$ | localization and abundance              | S288C+tag          | <a href="https://doi.org/10.1038/nmeth.3795">doi:10.1038/nmeth.3795</a>                                                  |
| Cut line. Rows above are the 151 that reach 200; rows below are the ranked reserve. |                                                             |                        |                                   |                                  |                     |                   |                                         |                    |                                                                                                                          |
| 152                                                                                 | Hénault 2023 (hybrid and introgression panel)               | Natural variation      | hybrids and introgression lines   | conditions                       | $5.0 \times 10^3$ ‡ | $5.0 \times 10^3$ | fitness                                 | isolate-WGS        | <a href="https://doi.org/10.1038/s41467-024-46155-5">doi:10.1038/s41467-024-46155-5</a>                                  |
| 153                                                                                 | Cubillos 2013 (F12 recombinant panel construction)          | Natural variation      | 192 sequenced F12 segregants      | conditions                       | $3.8 \times 10^3$ ‡ | $3.8 \times 10^3$ | growth traits                           | segregant-WGS      | <a href="https://doi.org/10.1534/genetics.113.155515">doi:10.1534/genetics.113.155515</a>                                |
| 154                                                                                 | Kessi-Perez 2019 (nitrogen-consumption QTL panel)           | Natural variation      | sequenced segregants / isolates   | nitrogen conditions              | 200‡                | 200               | nitrogen consumption, fermentation rate | segregant-WGS      | <a href="https://doi.org/10.1186/s13068-019-1440-9">doi:10.1186/s13068-019-1440-9</a>                                    |
| 155                                                                                 | Lian 2016 (combinatorial xylose pathway optimization)       | Metabolite / precursor | combinatorial pathway variants    | xylose                           | 100‡                | 100               | ethanol titer and rate on xylose        | engineered-chassis | <a href="https://doi.org/10.1016/j.ymben.2015.11.001">doi:10.1016/j.ymben.2015.11.001</a>                                |

**Table 2:** Sources for Table 1, in the same order. *Why* states what the row buys that the built set does not. *Data* is where the per-record values live; entries marked unconfirmed were not fetched live and must be checked before a loader is written. Every link is clickable.

| # | Dataset, citation and link                                                                                                                                                                                                        | Why                                                                                                                                                                                                                                                                                                                                                            | Data                                                                                                                            |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>Boocock 2025 (single-cell eQTL mapping)</b><br>Boocock J, Alexander N, Alamo Tapia L, Walter-McNeill L, Patel SP, Munugala C, Bloom JS, Kruglyak L. eLife 2025;13:e95566. doi:10.7554/eLife.95566                              | 27,744 single cells drawn from 393 sequenced recombinants in two carbon sources, a median of 17 cells per segregant. High-dimensional on both axes at once, which almost nothing in yeast is, and the closest existing template for the proposed design.                                                                                                       | mirrored as boocockSinglecellEQTLMapping2025; GEO (unconfirmed)                                                                 |
| 2 | <b>de Boer 2020 (100M random promoters)</b><br>de Boer CG, Vaishnav ED, Sadeh R, Abeyta EL, Friedman N, Regev A. Nat Biotechnol 2020;38:56-65. doi:10.1038/s41587-019-0315-8                                                      | The largest sequence-to-phenotype dataset in yeast by orders of magnitude, and every genotype is an exactly known 80mer. The perturbation space is continuous rather than a gene list, which is what a model of regulatory grammar needs and no deletion collection can supply.                                                                                | GEO (de Boer 2020); DREAM 2022 release of 6,739,258 promoters                                                                   |
| 3 | <b>Vaishnav 2022 (regulatory DNA fitness landscape)</b><br>Vaishnav ED, de Boer CG, Molinet J, Yassour M, Fan L, Adiconis X, Thompson DA, Levin JZ, Cubillos FA, Regev A. Nature 2022;603:455-463. doi:10.1038/s41586-022-04506-6 | Extends de Boer from measurement to a fitness landscape, and tests native and natural-isolate promoter variants against the random library. The bridge between synthetic sequence space and the natural variation the isolate panels carry.                                                                                                                    | GEO per the Nature data-availability statement (unconfirmed)                                                                    |
| 4 | <b>Lee 2014 (HIP-HOP fitness signatures)</b><br>Lee AY, St Onge RP, Proctor MJ, et al., Giaever G, Nislow C. Science 2014;344:208-211. doi:10.1126/science.1250217                                                                | Largest chemical-genetic matrix in existence, on the exact YKO genotype axis Costanzo and Kuzmin already use. Biggest single instance-count gain available. BLOCKED: WS15 already attempted it and it stands as awaiting author matrices, so the Science SI and the lab portal did not yield per-strain values. Unblocking is an author request, not a loader. | Science SI + Nislow/Giaever portal, neither of which yielded per-strain matrices; awaiting author matrices per the WS15 roadmap |
| 5 | <b>Albert 2018 (eQTL in 1,012 segregants)</b><br>Albert FW, Bloom JS, Siegel J, Day L, Kruglyak L. eLife 2018;7:e35471. doi:10.7554/eLife.35471                                                                                   | Transcriptomes on a thousand sequenced recombinants, so genotype maps to a genome-wide expression vector rather than one trait. The single best existing analog to a Perturb-seq readout on a high-dimensional genotype axis.                                                                                                                                  | GEO /eLife data availability (unconfirmed)                                                                                      |
| 6 | <b>Jakobson 2025 (genome-to-proteome map)</b><br>Jakobson CM, et al. Science 2025;390:eadu3198. doi:10.1126/science.adu3198                                                                                                       | Natural alleles mapped to enzyme abundance. A lower-risk engineering lever than a heterologous construct, and the proteome counterpart to the metabolite-QTL panels.                                                                                                                                                                                           | Science data-availability statement (accession unconfirmed)                                                                     |
| 7 | <b>Muenzner 2024 (natural-isolate proteome)</b><br>Muenzner J, et al., Ralser M. Nature 2024;630:149-157. doi:10.1038/s41586-024-07442-9                                                                                          | Proteomes on the sequenced isolate panel whose transcriptomes Caudal 2024 already supplies. That overlap is the paired anchor the RNA-to-protein inference thesis needs.                                                                                                                                                                                       | PRIDE PXD048219                                                                                                                 |
| 8 | <b>Hale 2024 (CRISPRi x natural variation)</b><br>Hale JJ, Matsui T, Goldstein I, et al., Kruglyak L. Nat Commun 2024;15:4234. doi:10.1038/s41467-024-48626-1                                                                     | Directly measures how a genetic perturbation's effect changes with genetic background. The single best dataset for asking whether a model trained on one background transfers to another.                                                                                                                                                                      | SRA PRJNA986287 + Figshare                                                                                                      |
| 9 | <b>Jackson 2020 (TF-deletion single-cell atlas)</b><br>Jackson CA, Castro DM, Saldi GA, Bonneau R, Gresham D. eLife 2020;9:e51254. doi:10.7554/eLife.51254                                                                        | Single-cell transcriptomes across a panel of regulator deletions in two media, used to learn a regulatory network. Few genotypes, but it is one of the only yeast datasets crossing a genetic perturbation with a per-cell transcriptome, which is the quadrant the Perturb-seq proposal targets.                                                              | GEO GSE125162 /GSE125163 (unconfirmed)                                                                                          |



Table 2, continued

| #  | Dataset, citation and link                                                                                                                                                                                                                                                                                  | Why                                                                                                                                                                                                                                                                           | Data                                                         |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| 10 | <b>Nguyen Ba 2022 (barcoded bulk QTL, 100k segregants)</b><br>Nguyen Ba AN, Lawrence KR, Rego-Costa A, Gopalakrishnan S, Temko D, Michor F, Desai MM. eLife 2022;11:e73983.<br><a href="https://doi.org/10.7554/eLife.73983">doi:10.7554/eLife.73983</a>                                                    | The largest genotyped recombinant panel in yeast, each segregant barcoded and low-coverage sequenced against known parents. Two orders of magnitude more genotypes than Bloom 2013 on the same kind of cross, which is what makes small-effect and epistatic loci detectable. | eLife data availability; SRA (unconfirmed)                   |
| 11 | <b>Galardini 2019 (four backgrounds x 38 conditions)</b><br>Galardini M, Busby BP, Vieitez C, Dunham AS, Typas A, Beltrao P. Mol Syst Biol 2019;15:e8831.<br><a href="https://doi.org/10.15252/msb.20198831">doi:10.15252/msb.20198831</a>                                                                  | The same knockout built independently in four sequenced backgrounds. Quantifies the 18.5 percent of deletion phenotypes that do not transfer, which is the generalization risk the whole substrate is exposed to.                                                             | GEO GSE123118 + github.com/mgalardini/2018koyeast            |
| 12 | <b>Bloom 2019 (16-parent cross)</b><br>Bloom JS, Boocock J, Treusch S, Sadhu MJ, Day L, Oates-Barker H, Kruglyak L. eLife 2019;8:e49212.<br><a href="https://doi.org/10.7554/eLife.49212">doi:10.7554/eLife.49212</a>                                                                                       | Largest sequenced recombinant panel in yeast. Sixteen founders means allelic diversity a two-parent cross cannot reach, and the genotypes are sequence, not markers.                                                                                                          | eLife SI + SRA (confirm accession before ingestion)          |
| 13 | <b>Parsons 2006 (bioactive-compound profiling)</b><br>Parsons AB, Lopez A, Givoni IE, et al., Boone C. Cell 2006;126:611-625.<br><a href="https://doi.org/10.1016/j.cell.2006.06.040">doi:10.1016/j.cell.2006.06.040</a>                                                                                    | Full-collection strain coverage against a modest compound panel, the complement to the diagnostic-panel screens above.                                                                                                                                                        | Cell SI tables (accession unconfirmed)                       |
| 14 | <b>Dutta 2026 (barcoded natural-isolate chemical response)</b><br>Dutta A, Garin M, Loegler V, Brach G, Friedrich A, Yoshimura M, Hirano H, Osada H, Boone C, Yashiroda Y, Hou J, Schacherer J. Nat Commun 2026.<br><a href="https://doi.org/10.1038/s41467-026-73532-z">doi:10.1038/s41467-026-73532-z</a> | Natural sequence diversity crossed with a chemogenomic panel, on isolates drawn from the sequenced 1,011 collection. The same strains already carry Caudal 2024 transcriptomes and Muenzner 2024 proteomes, so it is a three-modality join on one genotype axis.              | Nat Commun SI; isolates from the 1,011 panel (ENA)           |
| 15 | <b>Cooper 2010 (CE-MS amino-acid metabolome)</b><br>Cooper SJ, Finney GL, Brown SL, Nelson SI, Hesse J, MacCoss MJ, Fields S. Genome Res 2010;20:1288-1296.<br><a href="https://doi.org/10.1101/gr.105825.110">doi:10.1101/gr.105825.110</a>                                                                | An independent platform measuring the same trait class as the already-built Mulleder 2016 on an overlapping genotype axis. Two independent measurements of one phenotype is the cleanest available test of whether a model has learned biology or a batch.                    | Genome Research SI tables                                    |
| 16 | <b>Li 2016 (tRNA fitness landscape)</b><br>Li C, Qian W, Maclean CJ, Zhang J. Science 2016;352:837-840.<br><a href="https://doi.org/10.1126/science.aae0568">doi:10.1126/science.aae0568</a>                                                                                                                | A near-complete fitness landscape of one native yeast gene. Every genotype is a known sequence and the sampling is exhaustive rather than one allele per gene, which is the opposite of what a deletion collection provides.                                                  | Science SI (unconfirmed)                                     |
| 17 | <b>Aulakh 2025 (genome-scale ionome)</b><br>Aulakh SK, et al. Cell Syst 2025;16:101319.<br><a href="https://doi.org/10.1016/j.cels.2025.101319">doi:10.1016/j.cels.2025.101319</a>                                                                                                                          | The only gene-indexed ionome. Metal-cofactor supply gates iron-sulfur and zinc-dependent pathway enzymes, and no built dataset measures it.                                                                                                                                   | Cell Systems SI (accession unconfirmed)                      |
| 18 | <b>Peter 2018 (1,011 isolate genomes + phenome)</b><br>Peter J, De Chiara M, Friedrich A, et al., Schacherer J. Nature 2018;556:339-344.<br><a href="https://doi.org/10.1038/s41586-018-0030-5">doi:10.1038/s41586-018-0030-5</a>                                                                           | The genome backbone every other natural-variation row resolves against, and it carries its own phenotype panel. 1,625,809 high-quality SNPs at 232-fold mean coverage.                                                                                                        | ENA/SRA; phenotype tables via the paper and France Genomique |
| 19 | <b>Domingo 2018 (tRNA double-mutant landscape)</b><br>Domingo J, Diss G, Lehner B. Nature 2018;558:117-121.<br><a href="https://doi.org/10.1038/s41586-018-0170-7">doi:10.1038/s41586-018-0170-7</a>                                                                                                        | Pairwise epistasis at nucleotide resolution inside one gene. The higher-order structure Costanzo and Kuzmin map between genes, measured within a gene, which is the resolution inverse design actually needs.                                                                 | Nature SI (unconfirmed)                                      |



Table 2, continued

| #  | Dataset, citation and link                                                                                                                                                                                                                                                    | Why                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Data                                                                                                                                          |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| 20 | <b>Zhang 2021 (YETI titratable overexpression)</b><br>Zhang Y, Ho Yee Chow M, Ling Y, et al., Andrews BJ. Mol Syst Biol 2021;17:e10321.<br><a href="https://doi.org/10.15252/msb.202010321">doi:10.15252/msb.202010321</a>                                                    | Native-locus, continuously titratable overexpression. The dosage axis pathway engineering actually uses, and the only gain-of-function library that is not binary.                                                                                                                                                                                                                                                                                                    | Mol Syst Biol SI (repository accession unconfirmed)                                                                                           |
| 21 | <b>Peeters 2021 (fermentation-trait QTL atlas)</b><br>Peeters B, Reijbroek F, Verbaet J, et al., Jarosz DF, Verstrepen KJ. 2021. PMID 34727964.<br><a href="https://pubmed.ncbi.nlm.nih.gov/34727964/">PMID 34727964</a>                                                      | The only large sequenced panel with direct isobutanol, ethanol and glycerol titers. Hits the natural-isolate and the isobutanol asks at once.                                                                                                                                                                                                                                                                                                                         | Journal SI; raw reads at SRA/ENA (accession unconfirmed)                                                                                      |
| 22 | <b>Van Leeuwen 2024 (short-chain organic acids)</b><br>Shared and specific genetic determinants of tolerance to acetic, butyric and octanoic acid. PMC10903034.<br><a href="https://pubmed.ncbi.nlm.nih.gov/34727964/">PMC10903034</a>                                        | Three chemically homologous acids on one strain panel, so shared and acid-specific mechanisms separate within a single study rather than across incomparable screens.                                                                                                                                                                                                                                                                                                 | Journal SI (open-access PMC)                                                                                                                  |
| 23 | <b>Chica 2026 (AutoDRY autophagy screen)</b><br>Chica N, et al. Nat Cell Biol 2026;28:465-479.<br><a href="https://doi.org/10.1038/s41556-025-01837-0">doi:10.1038/s41556-025-01837-0</a>                                                                                     | Largest single-condition imaging screen with a live Dryad deposit, and it covers essential-adjacent genes via DAmP rather than stopping at the non-essential set.                                                                                                                                                                                                                                                                                                     | Dryad 10.5061/dryad.cfxpvnvxdh                                                                                                                |
| 24 | <b>Sopko 2006 (genome-scale overexpression toxicity)</b><br>Sopko R, Huang D, Preston N, et al., Boone C, Andrews B. Mol Cell 2006;21:319-330.<br><a href="https://doi.org/10.1016/j.molcel.2005.12.011">doi:10.1016/j.molcel.2005.12.011</a>                                 | 769 genes are growth-inhibitory when overexpressed. Overexpression toxicity is the constraint that caps titer when a pathway enzyme is pushed, and nothing in the built set measures it.                                                                                                                                                                                                                                                                              | Mol Cell SI tables                                                                                                                            |
| 25 | <b>Liu 2021 (tryptophan / isobutanol tolerance)</b><br>Liu H-L, Wang CH-T, Chiang EP-I, Huang C-C, Li W-H. Biotechnol Biofuels 2021;14:200.<br><a href="https://doi.org/10.1186/s13068-021-02048-z">doi:10.1186/s13068-021-02048-z</a>                                        | A second, independent full-collection isobutanol screen with a different readout. Paired with Kuroda it gives a same-phenotype, same-genotype, different-method replicate.                                                                                                                                                                                                                                                                                            | GEO GSE175794 (companion RNA-seq); screen in BMC SI                                                                                           |
| 26 | <b>Kuroda 2019 (isobutanol-specific tolerance)</b><br>Kuroda K, Hammer SK, Watanabe Y, Montano Lopez J, Fink GR, Stephanopoulos G, Ueda M, Avalos JL. Cell Syst 2019;9:534-547.e5.<br><a href="https://doi.org/10.1016/j.cels.2019.10.006">doi:10.1016/j.cels.2019.10.006</a> | The isobutanol screen, designed against an ethanol comparator so the alcohol-general response subtracts out. Its GLN3 hit raised production 4.9-fold, a rare tolerance-to-titer translation.                                                                                                                                                                                                                                                                          | ArrayExpress E-MTAB-8175 (RNA-seq); screen data in Cell Syst SI                                                                               |
| 27 | <b>Puddu 2019 (WGS of the deletion collection)</b><br>Puddu F, et al., Jackson SP. Nature 2019;573:416-420.<br><a href="https://doi.org/10.1038/s41586-019-1549-9">doi:10.1038/s41586-019-1549-9</a>                                                                          | Whole-genome sequence for every strain in the deletion collection. This is what turns the S288C-KO sequence basis from an assumption into a measurement, and it applies to roughly half the rows in this table at once.                                                                                                                                                                                                                                               | ENA/SRA (accession unconfirmed)                                                                                                               |
| 28 | <b>Turco 2023 (Yeast Phenome)</b><br>Turco G, Chang C, Wang RY, et al., Boone C, Andrews BJ, Roth FP. Sci Adv 2023;9:eadg5702.<br><a href="https://doi.org/10.1126/sciadv.adg5702">doi:10.1126/sciadv.adg5702</a>                                                             | The widest environment axis in yeast, 96 percent chemical. Held out of tier 1 because it is a meta-aggregation: its instance count re-serves primary screens, several already built, so the net-new fraction is unknown until it is de-duplicated against the rest of this table. A loader already exists and retains 49 growth screens, so this is in flight rather than net-new work; Lee 2014 is not among those screens, so YeastPhenome does not backfill row 1. | yeastphenome.org (Zenodo 10.5281/zenodo.7714347); loader torchcell/datasets/scerevisiae/yeastphenome.py, 49 screens, not yet in the built set |
| 29 | <b>N'Guessan 2025 (segregant scRNA-seq eQTL)</b><br>N'Guessan A, Boocock J, Kruglyak L, Albert FW. eLife 2025. PMC12303567.<br><a href="https://pubmed.ncbi.nlm.nih.gov/34727964/">PMC12303567</a>                                                                            | Single-cell transcriptomes indexed by recombinant natural genotype rather than by an engineered knockout. The closest existing analog to the proposed Perturb-seq design, and the only one that already pairs per-cell expression with sequenced genotypes at scale.                                                                                                                                                                                                  | eLife data availability; segregant panel from Bloom lineage                                                                                   |

Table 2, continued

| #  | Dataset, citation and link                                                                                                                                                                                                                    | Why                                                                                                                                                                                                                                   | Data                                                      |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| 30 | <b>Rossi 2021 (ChIP-exo protein architecture)</b><br>Rossi MJ, Kuntala PK, Lai WKM, et al., Pugh BF. Nature 2021;592:309-314.<br><a href="https://doi.org/10.1038/s41586-021-03314-8">doi:10.1038/s41586-021-03314-8</a>                      | Where roughly 400 regulatory proteins sit on the genome at near-base-pair resolution. This is the wiring prior that makes a regulatory-DNA model mechanistic instead of correlational, and it pairs directly with the MPRA rows.      | yeastepigenome.org; github.com/CEGRcode/2021-Rossi_Nature |
| 31 | <b>Piotrowski 2017 (MOSAIC diagnostic panel)</b><br>Piotrowski JS, Li SC, Deshpande R, et al., Boone C, Myers CL. Nat Chem Biol 2017;13:982-993.<br><a href="https://doi.org/10.1038/nchembio.2436">doi:10.1038/nchembio.2436</a>             | Highest compound count of any yeast screen. Held out of tier 1 on the genotype bar: 157 strains buys condition breadth, not genotype breadth, so it trains a compound embedding rather than a genotype model.                         | MOSAIC portal (mosaic.cs.umn.edu) + Nat Chem Biol SI      |
| 32 | <b>Hu 2007 (TF deletion expression compendium)</b><br>Hu Z, Killion PJ, Iyer VR. Nat Genet 2007;39:683-687.<br><a href="https://doi.org/10.1038/ng2012">doi:10.1038/ng2012</a>                                                                | Transcriptomes for the regulators specifically, predating and complementing the already-built Kemmeren compendium. Deleting a transcription factor is the perturbation whose transcriptome response is most interpretable.            | GEO (unconfirmed)                                         |
| 33 | <b>Bloom 2025 (global epistasis in a yeast cross)</b><br>Bloom JS, et al., Kruglyak L. 2025. PMID 40679398.<br><a href="https://pubmed.ncbi.nlm.nih.gov/40679398/">PMID 40679398</a>                                                          | Reports that many natural variants have effects that scale with background fitness, which is a specific, testable claim about how genotype effects compose. Directly relevant to whether a model trained on one background transfers. | journal SI (unconfirmed; recent)                          |
| 34 | <b>Gasch 2000 (environmental stress response)</b><br>Gasch AP, Spellman PT, Kao CM, et al., Brown PO. Mol Biol Cell 2000;11:4241-4257.<br><a href="https://doi.org/10.1091/mbc.11.12.4241">doi:10.1091/mbc.11.12.4241</a>                     | The canonical stress-response signature. Every genotype-indexed transcriptome in the built set needs it as the background any generic stress response is subtracted against.                                                          | SPELL /SGD Expression Connection; Stanford legacy mirror  |
| 35 | <b>Decourty 2021 (RNA-metabolism genetic interactions)</b><br>Decourty L, et al., Saveanu C. Nucleic Acids Res 2021;49:8535-8555.<br><a href="https://doi.org/10.1093/nar/gkab680">doi:10.1093/nar/gkab680</a>                                | Structurally identical to Costanzo and Kuzmin, so the existing loader pattern applies directly, and it extends the interaction network into RNA processing.                                                                           | NAR open-access SI (accession unconfirmed)                |
| 36 | <b>Leutert 2023 (phosphoproteome x 101 conditions)</b><br>Leutert M, et al., Villen J. Nat Struct Mol Biol 2023;30:1761-1773.<br><a href="https://doi.org/10.1038/s41594-023-01115-3">doi:10.1038/s41594-023-01115-3</a>                      | Post-translational enzyme control acts faster than transcription and is invisible to every other layer here. Condition axis only.                                                                                                     | PRIDE PXD034997                                           |
| 37 | <b>Cuperus 2017 (random 5' UTR library)</b><br>Cuperus JT, Groves B, Kuchina A, Rosenberg AB, Jovic N, Fields S, Seelig G. Genome Res 2017;27:2015-2024.<br><a href="https://doi.org/10.1101/gr.224964.117">doi:10.1101/gr.224964.117</a>     | The translational layer of the same idea at half a million exactly known sequences. Expression is set after transcription as well as at it, and nothing else here measures that axis.                                                 | GEO (unconfirmed)                                         |
| 38 | <b>Faure 2022 (doubledeepPCA)</b><br>Faure AJ, Domingo J, Schmiedel JM, Hidalgo-Carcedo C, Diss G, Lehner B. Nature 2022;604:175-183.<br><a href="https://doi.org/10.1038/s41586-022-04586-4">doi:10.1038/s41586-022-04586-4</a>              | Separates a variant's effect on folding from its effect on binding by measuring both in the same library. That decomposition is what turns a fitness landscape into a mechanistic one.                                                | Nature data availability (unconfirmed)                    |
| 39 | <b>Chong 2015 / CYCLOPs (single-cell proteome dynamics)</b><br>Chong YT, Koh JLY, Friesen H, et al., Andrews BJ, Boone C. Cell 2015;161:1413-1424.<br><a href="https://doi.org/10.1016/j.cell.2015.04.051">doi:10.1016/j.cell.2015.04.051</a> | Per-cell rather than per-strain-mean labels, over 20 million cells. The distributional readout a Perturb-seq design has to be scored against.                                                                                         | thecellvision.org/cyclops (bulk CSV)                      |

Table 2, continued

| #  | Dataset, citation and link                                                                                                                                                                                                                                                                                | Why                                                                                                                                                                                                                                                                                                                         | Data                                                    |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| 40 | <b>Brauer 2008 (growth-rate-controlled chemostat transcriptome)</b><br>Brauer MJ, Huttenhower C, Airoidi EM, et al., Botstein D. Mol Biol Cell 2008;19:352-367.<br><a href="https://doi.org/10.1091/mbc.e07-08-0779">doi:10.1091/mbc.e07-08-0779</a>                                                      | Separates growth rate from nutrient identity by design, which no batch-culture dataset can. A large share of any expression response is a growth-rate confound, and this is the reference for removing it.                                                                                                                  | GEO (unconfirmed)                                       |
| 41 | <b>Teyssonniere 2024 (species-wide trait survey)</b><br>Teyssonniere EM, Shichino Y, Mito M, Friedrich A, Iwasaki S, Schacherer J. PLoS Genet 2024;20:e1011119.<br><a href="https://doi.org/10.1371/journal.pgen.1011119">doi:10.1371/journal.pgen.1011119</a>                                            | Extends the phenotype axis of the sequenced panel far past Peter 2018's 36 conditions, and classifies traits by how monogenic or polygenic they are. That classification is a prior on which phenotypes a model can be expected to predict at all.                                                                          | PLoS Genetics SI (unconfirmed)                          |
| 42 | <b>Diss 2018 (Fos-Jun interaction double mutants)</b><br>Diss G, Lehner B. eLife 2018;7:e32472.<br><a href="https://doi.org/10.7554/eLife.32472">doi:10.7554/eLife.32472</a>                                                                                                                              | Exhaustive pairwise mutation of an interaction interface, read out in yeast. The genotype is a heterologous domain rather than a yeast gene, so it enters as an engineered chassis, but the epistasis structure is the deepest in the table.                                                                                | eLife data availability (unconfirmed)                   |
| 43 | <b>Warringer 2011 (trait variation across the SGRP panel)</b><br>Warringer J, Zorgo E, Cubillos FA, et al., Blomberg A, Liti G. PLoS Genet 2011;7:e1002111.<br><a href="https://doi.org/10.1371/journal.pgen.1002111">doi:10.1371/journal.pgen.1002111</a>                                                | A very wide condition panel on sequenced isolates, with growth-curve parameters rather than endpoint fitness. Pairs with Scan-o-matic as an actually-released panel rather than a platform.                                                                                                                                 | PLoS Genetics SI (unconfirmed)                          |
| 44 | <b>Dong 2021 (MAGIC + SAM biosensor)</b><br>Dong C, Schultz JC, Liu W, Lian J, Huang L, Xu Z, Zhao H. Metab Eng 2021;66:319-327.<br><a href="https://doi.org/10.1016/j.ymben.2021.03.005">doi:10.1016/j.ymben.2021.03.005</a>                                                                             | MAGIC re-run with a metabolite biosensor instead of a growth selection, so the label is product concentration rather than fitness. The pattern to copy for every precursor we care about, and the sibling of the already-built Lian 2019 furfural screen.                                                                   | Metab Eng SI (accession unconfirmed)                    |
| 45 | <b>Momen-Roknabadi 2020 (inducible CRISPRi library)</b><br>Momen-Roknabadi A, Oikonomou P, Zegans M, Tavazoie S. Commun Biol 2020;3:723.<br><a href="https://doi.org/10.1038/s42003-020-01452-9">doi:10.1038/s42003-020-01452-9</a>                                                                       | A second CRISPRi library on a different vector and guide-design rule. Whether a model trained on one library transfers to the other is a cheap, decisive test of guide-level overfitting. Not a Perturb-seq row: one edit per cell and a scalar fitness readout is low-dimensional on both axes, however large the library. | Addgene + Commun Biol SI                                |
| 46 | <b>Keren 2013 (promoter activity across conditions)</b><br>Keren L, Zackay O, Lotan-Pompan M, Barenholz U, Dekel E, Sasson V, Aidelberg G, Bren A, Zeevi D, Weinberger A, Alon U, Milo R, Segal E. Mol Syst Biol 2013;9:701.<br><a href="https://doi.org/10.1038/msb.2013.59">doi:10.1038/msb.2013.59</a> | Native promoter strength measured directly, across conditions and over time, rather than inferred from mRNA. This is the closest published answer to the question of how strong every promoter actually is.                                                                                                                 | Mol Syst Biol SI (unconfirmed)                          |
| 47 | <b>Hackett 2016 (SIMMER multi-omic flux)</b><br>Hackett SR, Zanotelli VRT, Xu W, et al., Rabinowitz JD. Science 2016;354:aaf2786.<br><a href="https://doi.org/10.1126/science.aaf2786">doi:10.1126/science.aaf2786</a>                                                                                    | The only jointly measured metabolome, proteome and fluxome. No genotype axis, so it is a mechanism prior rather than training data, but it is the only place the three layers are measured on the same cells.                                                                                                               | Science SI (accession unconfirmed)                      |
| 48 | <b>McGlinicy 2021 (genome-scale CRISPRi library)</b><br>McGlinicy NJ, Meacham ZA, Reynaud KK, Muller R, Baum R, Ingolia NT. BMC Genomics 2021;22:205.<br><a href="https://doi.org/10.1186/s12864-021-07518-0">doi:10.1186/s12864-021-07518-0</a>                                                          | Ten guides per gene gives a graded knockdown axis instead of a binary null, and it covers essential genes. This is the library a yeast Perturb-seq would most plausibly be built on.                                                                                                                                        | ingolia-lab.org/yeast-crispri + Addgene + BMC SI        |
| 49 | <b>Bao 2018 (CHAnGE single-nucleotide library)</b><br>Bao Z, Hamedirad M, Xue P, Xiao H, Tasan I, Chao R, Liang J, Zhao H. Nat Biotechnol 2018;36:505-508.<br><a href="https://doi.org/10.1038/nbt.4132">doi:10.1038/nbt.4132</a>                                                                         | Single-nucleotide resolution rather than whole-gene nulls. The only genotype axis in this table finer than an ORF, and the one that matches how a real strain-design edit is specified.                                                                                                                                     | Nat Biotechnol SI (raw sequencing location unconfirmed) |

Table 2, continued

| #  | Dataset, citation and link                                                                                                                                                                                                                        | Why                                                                                                                                                                                                                                                             | Data                                       |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| 50 | <b>Puchta 2016 (snoRNA U3 landscape)</b><br>Puchta O, Cseke B, Czaja H, Tollervey D, Sanguinetti G, Kudla G. Science 2016;352:840-844.<br><a href="https://doi.org/10.1126/science.aaf0965">doi:10.1126/science.aaf0965</a>                       | The companion landscape on a non-coding RNA, published alongside Li 2016. Two exhaustive landscapes on different molecule classes is a stronger test of a sequence model than either alone.                                                                     | Science SI (unconfirmed)                   |
| 51 | <b>Yoshikawa 2011 (deletion + overexpression phenome)</b><br>Yoshikawa K, Tanaka T, Furusawa C, Nagahisa K, Hirasawa T, Shimizu H. G3 2011;1:247-267.<br><a href="https://doi.org/10.1534/g3.111.000695">doi:10.1534/g3.111.000695</a>            | The same genes phenotyped in both loss- and gain-of-function side by side. A matched over/under pair on one platform is what a dosage-aware model needs and nothing else here provides.                                                                         | G3 open-access SI                          |
| 52 | <b>Ho 2009 (MoBY-ORF barcoded overexpression)</b><br>Ho CH, Magtanong L, Barker SL, et al., Boone C. Nat Biotechnol 2009;27:369-377.<br><a href="https://doi.org/10.1038/nbt.1534">doi:10.1038/nbt.1534</a>                                       | Native-promoter overexpression against a compound panel, giving a third dosage state alongside deletion and knockdown on the same chemogenomic readout.                                                                                                         | moby.ccbr.utoronto.ca + Andrews lab portal |
| 53 | <b>Bloom 2013 (1,008 BYxRM segregants)</b><br>Bloom JS, Ehrenreich IM, Loo WT, Lite TL, Kruglyak L. Nature 2013;494:234-237.<br><a href="https://doi.org/10.1038/nature11867">doi:10.1038/nature11867</a>                                         | The reference two-parent panel the field calibrates against, and the direct predecessor of Bloom 2019 and Nguyen Ba 2022. Ingesting all three gives one cross at three panel sizes, which is a clean test of how much genotype coverage a model actually needs. | Nature SI                                  |
| 54 | <b>Zhu 2014 (kinase / phosphatase lipidomics)</b><br>Zhu Z, Loewen CJR. Mol Biol Cell 2014. PMC4196872.<br><a href="https://pubmed.ncbi.nlm.nih.gov/24196872/">PMC4196872</a>                                                                     | A gene-indexed lipidome on a regulatory gene set. Complements the already-built da Silveira 2014 lipidomics, which has no kinase axis, and reads out the malonyl-CoA sink directly.                                                                             | Mol Biol Cell SI Table S4 (Excel)          |
| 55 | <b>Xiao 2014 (genome-wide RNAi furfural tolerance)</b><br>Xiao H, Zhao H. Biotechnol Biofuels 2014;7:78.<br><a href="https://doi.org/10.1186/1754-6834-7-78">doi:10.1186/1754-6834-7-78</a>                                                       | Furfural via a knockdown rather than a deletion axis, so it reaches essential genes and gives a dose-graded rather than binary perturbation on the flagship hydrolysate inhibitor.                                                                              | BMC open-access SI                         |
| 56 | <b>Crook 2016 (tunable RNAi, isobutanol + 1-butanol)</b><br>Crook N, Sun J, Morse N, Schmitz A, Alper HS. Appl Microbiol Biotechnol 2016;100:10005-10018. PMID 27654654.<br><a href="https://pubmed.ncbi.nlm.nih.gov/27654654/">PMID 27654654</a> | Two related alcohols at graded knockdown strength. The dosage axis is the signal a binary screen throws away, and Hsp70 emerged only because of it.                                                                                                             | AMB SI (DOI is an open verification item)  |
| 57 | <b>Douglas 2012 (barcoded overexpression fitness)</b><br>Douglas AC, Smith AM, Sharifpoor S, et al., Andrews BJ, Boone C, Nislow C. G3 2012;2:1279-1289.<br><a href="https://doi.org/10.1534/g3.112.003400">doi:10.1534/g3.112.003400</a>         | The same barcode-sequencing pipeline as the already-built Hillenmeyer FitDb, run in the opposite dosage direction. Lowest harmonization cost of any overexpression row.                                                                                         | G3 open-access SI                          |
| 58 | <b>She 2018 (natural variant effects across a large isolate panel)</b><br>She R, Jarosz DF. Cell 2018;172:478-490.<br><a href="https://doi.org/10.1016/j.cell.2017.12.015">doi:10.1016/j.cell.2017.12.015</a>                                     | Maps how often a natural variant's effect depends on the strain it sits in, across a wide condition panel. The natural-variation counterpart to Galardini's four-background deletion tensor.                                                                    | Cell SI (unconfirmed)                      |
| 59 | <b>Costello 2020 (bioreactor Bar-seq)</b><br>Costello Z, Wehrs M, Mukhopadhyay A, et al. Microb Cell Fact 2020;19:167.<br><a href="https://doi.org/10.1186/s12934-020-01423-z">doi:10.1186/s12934-020-01423-z</a>                                 | The only fitness measurement at bioreactor scale. Shake-flask ranking is known to diverge from fed-batch, and nothing else here tests that.                                                                                                                     | Microb Cell Fact open-access SI            |
| 60 | <b>Roy 2018 (multiplexed precision editing)</b><br>Roy KR, Smith JD, Vonesch SC, et al., St Onge RP. Nat Biotechnol 2018;36:512-520.<br><a href="https://doi.org/10.1038/nbt.4137">doi:10.1038/nbt.4137</a>                                       | Genomic rather than plasmid barcodes, which removes the barcode-swapping artifact that quietly corrupts pooled fitness data. The methodological complement to CHAnGE.                                                                                           | Nat Biotechnol SI (accession unconfirmed)  |

Table 2, continued

| #  | Dataset, citation and link                                                                                                                                                                                                                        | Why                                                                                                                                                                                                                                                                                                                                                                                                 | Data                                              |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| 61 | <b>Gerke 2017 (urea-cycle mQTL)</b><br>Gerke J, et al., Fay JC. Genetics 2017;206:2199.<br><a href="https://doi.org/10.1534/genetics.117.201107">doi:10.1534/genetics.117.201107</a>                                                              | A second mQTL panel from a different cross (oak x wine), so cross-background generalization of metabolite QTL can be tested rather than assumed.                                                                                                                                                                                                                                                    | Genetics (GSA) SI                                 |
| 62 | <b>Gurvitz (fatty-acid utilization screen)</b><br>Gurvitz A, et al. Mol Syst Biol (EMBO Press).<br><a href="https://embopress.org/doi/pdf/10.1038/msb4100051">embopress.org/doi/pdf/10.1038/msb4100051</a>                                        | Whole-collection coverage against two fatty-acid substrates. Relevant to the lipid and malonyl-CoA branch, at the cost of manual extraction from an embedded table.                                                                                                                                                                                                                                 | Mol Syst Biol Table I (PDF-embedded)              |
| 63 | <b>Mira 2010 (acetic acid tolerance, full collection)</b><br>Mira NP, Palma M, Guerreiro JF, Sa-Correia I. Microb Cell Fact 2010;9:79.<br><a href="https://doi.org/10.1186/1475-2859-9-79">doi:10.1186/1475-2859-9-79</a>                         | About 650 tolerance determinants on the full deletion collection at controlled pH. The loss-of-function counterpart to the already-built Mormino biosensor screen and to Mukherjee's essential-gene CRISPRi, so the same phenotype is covered by deletion, knockdown and biosensor.                                                                                                                 | Microb Cell Fact SI (PMC2972246)                  |
| 64 | <b>Gameiro 2025 (AID2 degron libraries)</b><br>Gameiro E, Juarez-Nunez KA, Fung JJ, Shankar S, Luke B, Khmelinskii A. J Cell Biol 2025;224:e202409007.<br><a href="https://doi.org/10.1083/jcb.202409007">doi:10.1083/jcb.202409007</a>           | Acute depletion on a minutes-to-hours timescale, covering essential genes without the pre-adaptation a deletion strain has already done.                                                                                                                                                                                                                                                            | J Cell Biol SI; strains via IMB Mainz / EUROSCARF |
| 65 | <b>Si 2017 (RAGE RNAi, isobutanol titer)</b><br>Si T, Chao R, Min Y, Wu Y, Ren W, Zhao H. Nat Commun 2017;8:15187.<br><a href="https://doi.org/10.1038/ncomms15187">doi:10.1038/ncomms15187</a>                                                   | One of very few library screens whose endpoint is a titer in g/L rather than a fitness proxy, and it is iterative, so it carries combinatorial rather than single-locus effects.                                                                                                                                                                                                                    | Nat Commun SI (accession unconfirmed)             |
| 66 | <b>Hamedirad 2018 (RAGE xylose utilization)</b><br>Hamedirad M, Lian J, Li H, Zhao H. Biotechnol Bioeng 2018;115:1552-1560.<br><a href="https://doi.org/10.1002/bit.26570">doi:10.1002/bit.26570</a>                                              | Pentose utilization is the other half of the recalcitrant-biomass problem, and no built dataset covers a xylose selection.                                                                                                                                                                                                                                                                          | Biotechnol Bioeng SI (accession unconfirmed)      |
| 67 | <b>Newman 2006 (protein noise, GFP library)</b><br>Newman JRS, Ghaemmaghani S, Ihmels J, Breslow DK, Noble M, DeRisi JL, Weissman JS. Nature 2006;441:840-846.<br><a href="https://doi.org/10.1038/nature04785">doi:10.1038/nature04785</a>       | Measures the distribution rather than the mean, per protein, in two media. Noise is the quantity a per-cell readout recovers and a bulk one destroys, so this sets what a single-cell design should expect to see.                                                                                                                                                                                  | Nature SI                                         |
| 68 | <b>Pereira 2014 (wheat-straw hydrolysate)</b><br>Pereira FB, Guimaraes PMR, Gomes DG, et al., Domingues L. J Ind Microbiol Biotechnol 2014;41:1753-1761.<br><a href="https://doi.org/10.1007/s10295-014-1519-z">doi:10.1007/s10295-014-1519-z</a> | Real industrial wheat-straw hydrolysate rather than reconstituted single toxins. The recalcitrant-biomass condition as a process actually presents it.                                                                                                                                                                                                                                              | Journal SI                                        |
| 69 | <b>Endo 2008 (vanillin tolerance)</b><br>Endo A, Nakamura T, Ando A, Tokuyasu K, Shima J. Appl Environ Microbiol 2008;74:7175-7185.<br><a href="https://doi.org/10.1128/AEM.01541-08">doi:10.1128/AEM.01541-08</a>                                | Vanillin is the one major hydrolysate phenolic with no dedicated screen in the built set; it is currently only folded into broad compound panels.                                                                                                                                                                                                                                                   | AEM SI (PMC2375868)                               |
| 70 | <b>Sharifpoor 2012 (kinome synthetic dosage lethality)</b><br>Sharifpoor S, van Dyk D, Costanzo M, et al., Boone C, Andrews BJ. Genome Res 2012;22:791-801.<br><a href="https://doi.org/10.1101/gr.129213.111">doi:10.1101/gr.129213.111</a>      | The overexpression-by-deletion axis, which neither the built SynLethDB nor Costanzo and Kuzmin carries: SynLethDB models only synthetic lethality and synthetic rescue, both two-loss-of-function, so synthetic dosage lethality is out of its scope by construction and this paper's PMID is absent from both its layers. Kinases are also the regulatory nodes flux redirection actually targets. | Genome Research SI                                |
| 71 | <b>Ambroset 2014 (metabolite QTL)</b><br>Ambroset C, et al., Fay JC. PLoS Genet 2014;10:e1004142.<br><a href="https://doi.org/10.1371/journal.pgen.1004142">doi:10.1371/journal.pgen.1004142</a>                                                  | The founding metabolite-QTL panel. Pairs with Cooper and Mulleder as the natural-variation view of a phenotype space the deletion collection covers by engineering.                                                                                                                                                                                                                                 | PLOS Genetics SI                                  |

Table 2, continued

| # Dataset, citation and link                                                                                                                                                                                                                                               | Why                                                                                                                                                                                                                                                                                    | Data                                             |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| 72 <b>Mavor 2016 (ubiquitin DMS across conditions)</b><br>Mavor D, Barlow K, Thompson S, et al., Bolon DNA, Fraser JS. eLife 2016;5:e15802.<br><a href="https://doi.org/10.7554/eLife.15802">doi:10.7554/eLife.15802</a>                                                   | A complete single-mutant scan of an essential yeast gene, repeated across stresses. Variant effects are conditional, and this is the cleanest demonstration of that in the table.                                                                                                      | eLife data availability (unconfirmed)            |
| 73 <b>Blank 2005 (13C metabolic flux)</b><br>Blank LM, Kuepfer L, Sauer U. Genome Biol 2005;6:R49.<br><a href="https://doi.org/10.1186/gb-2005-6-6-r49">doi:10.1186/gb-2005-6-6-r49</a>                                                                                    | The only real flux measurement tied to single-gene deletions. Everything else in the built set is a concentration or a titer, which is a proxy for flux rather than flux.                                                                                                              | Genome Biology open-access SI                    |
| 74 <b>Flynn 2020 (Hsp90 DMS across environments)</b><br>Flynn JM, Rossouw A, Cote-Hammarlof P, et al., Bolon DNA. eLife 2020;9:e53810.<br><a href="https://doi.org/10.7554/eLife.53810">doi:10.7554/eLife.53810</a>                                                        | A chaperone hub scanned under multiple stresses, so the environment-dependence of a variant effect is measured rather than assumed. Hsp90 buffers exactly the variation the natural-isolate rows carry.                                                                                | eLife data availability (unconfirmed)            |
| 75 <b>Ho 2018 (unified absolute protein abundance)</b><br>Ho B, Baryshnikova A, Brown GW. Cell Syst 2018;6:192-205.<br><a href="https://doi.org/10.1016/j.cels.2017.12.004">doi:10.1016/j.cels.2017.12.004</a>                                                             | Reconciles 21 independent abundance datasets into one unified estimate per protein. A better absolute anchor than any single study, including Ghaemmaghmi, and the reference frame every relative proteomics row needs.                                                                | Cell Systems SI                                  |
| 76 <b>Renganaath 2020 (natural promoter-variant MPRA)</b><br>Renganaath K, Cheung R, Day L, Kosuri S, Kruglyak L, Albert FW. eLife 2020;9:e62669.<br><a href="https://doi.org/10.7554/eLife.62669">doi:10.7554/eLife.62669</a>                                             | Native promoter variation measured allele by allele rather than sampled randomly, with 451 variants called causal. The row that joins the synthetic MPRA libraries to the natural-isolate panels: same readout, real alleles.                                                          | eLife data availability; GEO (unconfirmed)       |
| 77 <b>Chang 2012 (organic-acid production screen)</b><br>Chang HJ, Suga H, et al. 2012. PMID 22277779.<br><a href="https://pubmed.ncbi.nlm.nih.gov/22277779/">PMID 22277779</a>                                                                                            | A whole-collection screen whose label is production rather than fitness. Coarse and PDF-only, but organic-acid overproduction at collection scale exists nowhere else.                                                                                                                 | PDF tables only; manual transcription required   |
| 78 <b>Trikka 2015 (carotenogenic heterozygous screen)</b><br>Trikka FA, Nikolaidis A, Athanasakoglou A, et al., Kampranis SC, Makris AM. Microb Cell Fact 2015;14:60.<br><a href="https://doi.org/10.1186/s12934-015-0246-0">doi:10.1186/s12934-015-0246-0</a>             | The second carotenoid deletion-collection screen, and the only haploinsufficiency design here. Halving dosage finds genes a null would kill, which is where the isoprenoid flux control sits. Currently listed as figure-only, so confirm Additional file 1 carries per-strain scores. | BMC Additional files 1-2                         |
| 79 <b>Mukherjee 2021 (CRISPRi essential genes x acetic acid)</b><br>Mukherjee V, Lind U, St Onge RP, Blomberg A, Nystrom T. mSystems 2021;6:e00410-21.<br><a href="https://doi.org/10.1128/mSystems.00410-21">doi:10.1128/mSystems.00410-21</a>                            | Acetic acid is the dominant hydrolysate stressor, and this reaches the essential genes a deletion collection structurally cannot.                                                                                                                                                      | mSystems open-access SI                          |
| 80 <b>Sardi 2018 (natural variation in hydrolysate tolerance)</b><br>Sardi M, Paithane V, Place M, Robinson DE, Hose J, Wohlbach DJ, Gasch AP. PLoS Genet 2018;14:e1007217.<br><a href="https://doi.org/10.1371/journal.pgen.1007217">doi:10.1371/journal.pgen.1007217</a> | Natural variation in exactly the bioprocess phenotype the deletion screens cover by engineering. Pairs with Vanacloig-Pedros and Pereira to ask whether tolerance loci found by deletion recur as natural alleles.                                                                     | PLoS Genetics SI (unconfirmed)                   |
| 81 <b>Breslow 2008 (DAmP hypomorphic collection)</b><br>Breslow DK, Cameron DM, Collins SR, et al., Weissman JS. Nat Methods 2008;5:711-718.<br><a href="https://doi.org/10.1038/nmeth.1234">doi:10.1038/nmeth.1234</a>                                                    | Partial loss of function for essential genes at the native locus, about 82 percent of the essential genome. The built set has no essential-gene hypomorph axis at all.                                                                                                                 | Nat Methods SI; Horizon YSC5050/5090/5093/5094   |
| 82 <b>Kofoed 2015 (barcoded ts alleles)</b><br>Kofoed M, Milbury KL, Chiang JH, et al., Hieter P, Stirling PC. G3 2015;5:1879-1887.<br><a href="https://doi.org/10.1534/g3.115.019174">doi:10.1534/g3.115.019174</a>                                                       | Conditional essential-gene loss of function as point mutants at native loci, barcoded so it runs through the same pooled pipeline as the deletion collection.                                                                                                                          | G3 open-access SI; strains via Dharmacon/Horizon |



Table 2, continued

| #  | Dataset, citation and link                                                                                                                                                                                                                                      | Why                                                                                                                                                                                                                                                         | Data                                                                     |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| 83 | <b>Lian 2017 (CRISPR-AID, beta-carotene)</b><br>Lian J, Hamedirad M, Hu S, Zhao H. Nat Commun 2017;8:1688.<br><a href="https://doi.org/10.1038/s41467-017-01695-x">doi:10.1038/s41467-017-01695-x</a>                                                           | The tri-functional chassis MAGIC was built on, read out on beta-carotene. Small, but it is a combinatorial genotype with a measured product titer, which is the exact record type inverse strain design needs.                                              | Nat Commun SI                                                            |
| 84 | <b>Zhang 2022 (GCE-SCRaMbLE recombination outcomes)</b><br>Zhang W, Lazar-Stefanita L, Yamashita H, et al., Boeke JD. Nat Commun 2022;13:5836.<br><a href="https://doi.org/10.1038/s41467-022-33606-0">doi:10.1038/s41467-022-33606-0</a>                       | Each cell carries tens of simultaneous structural edits, sequenced afterwards. The deepest per-cell perturbation in the table: a genotype here is a rearranged chromosome, not an allele, which is what inverse strain design eventually has to predict on. | mirrored as<br>zhangSystematicDissectionKey2022; SRA (unconfirmed)       |
| 85 | <b>Cubillos 2017 (nitrogen-consumption QTL)</b><br>Cubillos FA, Brice C, Molinet J, et al., Martinez C. G3 2017;7:1693-1705.<br><a href="https://doi.org/10.1534/g3.117.042127">doi:10.1534/g3.117.042127</a>                                                   | One of the few QTL panels whose genotype location is confirmed rather than assumed, and nitrogen assimilation sets higher-alcohol and ester formation.                                                                                                      | BioProject PRJNA379146 (RNA-seq);<br>genotypes in Cubillos 2013 Table S1 |
| 86 | <b>Hillenmeyer 2010 (systematic fitness reanalysis)</b><br>Hillenmeyer ME, Ericson E, Davis RW, Nislow C, Koller D, Giaever G. Genome Biol 2010;11:R30.<br><a href="https://doi.org/10.1186/gb-2010-11-3-r30">doi:10.1186/gb-2010-11-3-r30</a>                  | The reprocessed and normalized form of the FitDb compendium, with per-condition significance. Ingesting the reanalysis rather than the raw release avoids redoing the normalization the authors already validated.                                          | Genome Biology SI (unconfirmed)                                          |
| 87 | <b>Smith 2008 (gene-environment interaction eQTL)</b><br>Smith EN, Kruglyak L. PLoS Biol 2008;6:e83.<br><a href="https://doi.org/10.1371/journal.pbio.0060083">doi:10.1371/journal.pbio.0060083</a>                                                             | Transcriptomes on the same segregants in two carbon sources, so an expression QTL can be shown to be condition-dependent. Small, and it is the design Boocock 2025 scales up.                                                                               | GEO (unconfirmed)                                                        |
| 88 | <b>Lenstra 2011 (chromatin regulator deletion expression)</b><br>Lenstra TL, Benschop JJ, Kim T, et al., Holstege FCP. Mol Cell 2011;42:536-549.<br><a href="https://doi.org/10.1016/j.molcel.2011.03.026">doi:10.1016/j.molcel.2011.03.026</a>                 | Transcriptomes for the chromatin machinery specifically, from the group behind the built Kemmeren compendium and in the same format. Chromatin regulators are the class whose deletion most reshapes the expression landscape.                              | GEO (unconfirmed)                                                        |
| 89 | <b>Braberg 2020 (point-mutant E-MAP)</b><br>Braberg H, et al., Krogan NJ. Science 2020;370:eaaz4910.<br><a href="https://doi.org/10.1126/science.aaz4910">doi:10.1126/science.aaz4910</a>                                                                       | Sub-gene resolution genetic interactions. Narrow in gene scope, but it is the only demonstration that the interaction formalism extends below the ORF.                                                                                                      | Science SI (accession unconfirmed)                                       |
| 90 | <b>Breker 2013 (GFP library under stress)</b><br>Breker M, Gymrek M, Schuldiner M. J Cell Biol 2013;200:839-850.<br><a href="https://doi.org/10.1083/jcb.201301120">doi:10.1083/jcb.201301120</a>                                                               | The GFP collection re-imaged under stress rather than at baseline, so relocalization is measured as a response. Feeds LoQAtE, and adds the condition axis Huh 2003 lacks.                                                                                   | LoQAtE (weizmann.ac.il/molgen/loqate)                                    |
| 91 | <b>Tkach 2012 (protein relocalization under DNA damage)</b><br>Tkach JM, Yimit A, Lee AY, et al., Andrews BJ, Brown GW. Nat Cell Biol 2012;14:966-976.<br><a href="https://doi.org/10.1038/ncb2549">doi:10.1038/ncb2549</a>                                     | The GFP collection imaged under damage, giving a second condition axis on the imaging layer alongside Breker's stress panel. Relocalization is a response no abundance measurement captures.                                                                | TheCellVision /Nat Cell Biol SI (unconfirmed)                            |
| 92 | <b>Ehrenreich 2010 (X-QTL bulk segregant mapping)</b><br>Ehrenreich IM, Torabi N, Jia Y, Kent J, Martis S, Shapiro JA, Gresham D, Caudy AA, Kruglyak L. Nature 2010;464:1039-1042.<br><a href="https://doi.org/10.1038/nature08923">doi:10.1038/nature08923</a> | Millions of segregants assayed in bulk rather than individually, so power comes from pool size instead of strain construction. The phenotype is per-pool rather than per-strain, which is a different record type and needs a pooled-genotype schema.       | Nature SI (unconfirmed)                                                  |
| 93 | <b>Skelly 2013 (expression variation across isolates)</b><br>Skelly DA, Merrihew GE, Riffle M, et al., MacCoss MJ, Akey JM. Genome Res 2013;23:1496-1504.<br><a href="https://doi.org/10.1101/gr.155762.113">doi:10.1101/gr.155762.113</a>                      | Transcriptome and proteome on the same isolates, which is the paired anchor the RNA-to-protein inference argument needs. Few strains, but the pairing is measured rather than joined across studies.                                                        | Genome Research SI (unconfirmed)                                         |



Table 2, continued

| #   | Dataset, citation and link                                                                                                                                                                                                                              | Why                                                                                                                                                                                                                            | Data                                  |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| 94  | <b>Denervaud 2013 (microfluidic GFP dynamics)</b><br>Denervaud N, Becker J, Delgado-Gonzalo R, et al., Maerkl SJ. PNAS 2013;110:15842-15847.<br><a href="https://doi.org/10.1073/pnas.1308265110">doi:10.1073/pnas.1308265110</a>                       | Single-cell protein dynamics under controlled environmental switches, rather than endpoint snapshots. The temporal axis every other imaging row lacks.                                                                         | PNAS SI (unconfirmed)                 |
| 95  | <b>Brogaard 2012 (chemical nucleosome map)</b><br>Brogaard K, Xi L, Wang JP, Widom J. Nature 2012;486:496-501.<br><a href="https://doi.org/10.1038/nature11142">doi:10.1038/nature11142</a>                                                             | Nucleosome positioning at base-pair resolution genome-wide. Promoter accessibility is what the MPRA rows are implicitly measuring, and this is the structural map underneath it.                                               | GEO (unconfirmed)                     |
| 96  | <b>Xia 2022 (proteome allocation across conditions)</b><br>Xia J, Sanchez BJ, Chen Y, Campbell K, Kasvandik S, Nielsen J. Nat Commun 2022;13:2819.<br><a href="https://doi.org/10.1038/s41467-022-30513-2">doi:10.1038/s41467-022-30513-2</a>           | Proteome allocation measured across a designed condition grid and used to parameterize enzyme-constrained models. The bridge between this table's proteomics rows and the Yeast9 modeling scaffold.                            | Nat Commun SI (unconfirmed)           |
| 97  | <b>Melamed 2013 (PAB1 RRM domain scan)</b><br>Melamed D, Young DL, Gamble CE, Miller CR, Fields S. RNA 2013;19:1537-1551.<br><a href="https://doi.org/10.1261/rna.040709.113">doi:10.1261/rna.040709.113</a>                                            | Dense coverage of one RNA-binding domain including double mutants, on a native essential yeast gene.                                                                                                                           | journal SI (unconfirmed)              |
| 98  | <b>De Chiara 2022 (domestication and genome instability)</b><br>De Chiara M, Barre BP, Persson K, et al., Liti G. Nat Ecol Evol 2022;6:761-773.<br><a href="https://doi.org/10.1038/s41559-022-01671-9">doi:10.1038/s41559-022-01671-9</a>              | Ties genome instability and aneuploidy in the isolate panel to phenotype. Industrial strains acquire aneuploidy under selection, so this is the natural-variation view of a failure mode long fermentations create.            | Nat Ecol Evol SI (unconfirmed)        |
| 99  | <b>Guo 2018 (CRISPR-Cas9 tiling of essential genes)</b><br>Guo X, Chavez A, Tung A, et al., Church GM, Shalem O. Nat Biotechnol 2018;36:540-546.<br><a href="https://doi.org/10.1038/nbt.4147">doi:10.1038/nbt.4147</a>                                 | Tiles guides across genes rather than sampling a few per gene, so the readout resolves within-gene functional regions. A finer input axis than a gene-level library.                                                           | Nat Biotechnol SI (unconfirmed)       |
| 100 | <b>Yvert 2013 (single-cell trait variation across isolates)</b><br>Yvert G, Ohnuki S, Nogami S, Imanaga Y, Fehrman S, Schacherer J, Ohya Y. BMC Syst Biol 2013;7:54.<br><a href="https://doi.org/10.1186/1752-0509-7-54">doi:10.1186/1752-0509-7-54</a> | CalMorph morphology on natural isolates rather than deletion strains, in the same feature space as the already-built Ohya and Ohnuki datasets. Joins the morphology layer to the natural-variation axis without harmonization. | BMC SI (unconfirmed)                  |
| 101 | <b>Lahtvee 2017 (absolute proteome across conditions)</b><br>Lahtvee PJ, Sanchez BJ, Smialowska A, et al., Nielsen J. Cell Syst 2017;4:495-504.<br><a href="https://doi.org/10.1016/j.cels.2017.03.003">doi:10.1016/j.cels.2017.03.003</a>              | Absolute enzyme copy numbers across conditions, which is the quantity enzyme-constrained metabolic models consume. Ties this table's proteomics to the Yeast9 modeling scaffold.                                               | Cell Systems SI (unconfirmed)         |
| 102 | <b>Weill 2018 (SWAT libraries)</b><br>Weill U, Yofe I, Sass E, et al., Schuldiner M. Nat Methods 2018;15:617-622.<br><a href="https://doi.org/10.1038/s41592-018-0044-9">doi:10.1038/s41592-018-0044-9</a>                                              | The library the degron rows are built from. Ingest for its abundance and localization tables; its main value is as the cassette definition those rows' sequences resolve against.                                              | Nat Methods SI; strains via EUROSCARF |
| 103 | <b>Kinsler 2020 (barcoded pleiotropy across environments)</b><br>Kinsler G, Geiler-Samerotte K, Petrov DA. eLife 2020;9:e61271.<br><a href="https://doi.org/10.7554/eLife.61271">doi:10.7554/eLife.61271</a>                                            | Sequenced adaptive clones assayed across dozens of environments, so one genotype yields a fitness vector rather than a point. Included because the clones were sequenced; the parent barcoding study was not, and is excluded. | eLife data availability (unconfirmed) |
| 104 | <b>Nadal-Ribelles 2019 (sensitive yeast scRNA-seq)</b><br>Nadal-Ribelles M, Islam S, Wei W, et al., Posas F, Steinmetz LM. Nat Microbiol 2019;4:683-692.<br><a href="https://doi.org/10.1038/s41564-018-0346-9">doi:10.1038/s41564-018-0346-9</a>       | Establishes within-clone transcript correlation, which is the noise floor any per-strain Perturb-seq mean has to clear. Directly sizes the cells-per-guide question.                                                           | Nat Microbiol SI /GEO                 |

Table 2, continued

| #   | Dataset, citation and link                                                                                                                                                                                                                                                               | Why                                                                                                                                                                                                                                                          | Data                                               |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| 105 | <b>Gasch 2017 (single-cell stress heterogeneity)</b><br>Gasch AP, Yu FB, Hose J, et al., Quake SR. PLoS Biol 2017;15:e2004050.<br><a href="https://doi.org/10.1371/journal.pbio.2004050">doi:10.1371/journal.pbio.2004050</a>                                                            | Separates intrinsic from extrinsic expression heterogeneity under stress. Sets what fraction of Perturb-seq variance is recoverable signal rather than cell-state noise.                                                                                     | PLoS Biology SI /GEO                               |
| 106 | <b>Pelechano 2013 (transcript isoform landscape)</b><br>Pelechano V, Wei W, Steinmetz LM. Nature 2013;497:127-131.<br><a href="https://doi.org/10.1038/nature12121">doi:10.1038/nature12121</a>                                                                                          | Shows a single gene produces many transcript isoforms with distinct 5' and 3' ends. Every expression row in this table collapses that to one number per gene, and this is the map of what that collapse discards.                                            | ArrayExpress /GEO (unconfirmed)                    |
| 107 | <b>Deutschbauer 2005 (haploinsufficiency, genome-wide)</b><br>Deutschbauer AM, Jaramillo DF, Proctor M, Kumm J, Hillenmeyer ME, Davis RW, Nislow C, Giaever G. Genetics 2005;169:1915-1925.<br><a href="https://doi.org/10.1534/genetics.104.036871">doi:10.1534/genetics.104.036871</a> | Gene-dosage sensitivity genome-wide, from halving rather than removing. Dosage-sensitive genes are exactly the ones whose overexpression in a pathway will misbehave.                                                                                        | Genetics SI (unconfirmed)                          |
| 108 | <b>Valenti 2025 (SWAT degron + GFP collection)</b><br>Valenti M, et al. J Cell Biol 2025;224:e202409050.<br><a href="https://doi.org/10.1083/jcb.202409050">doi:10.1083/jcb.202409050</a>                                                                                                | Depletion and its abundance consequence measured in the same strain, so enzyme dosage and its phenotype are not joined across two experiments.                                                                                                               | J Cell Biol SI                                     |
| 109 | <b>Sousa 2013 (acetate resistance overexpression screen)</b><br>Sousa M, Duarte AM, Fernandes TR, et al. 2013. PMID 23262128.<br>PMID 23262128                                                                                                                                           | Gain-of-function on the same stress the deletion and CRISPRi rows cover by loss. Over and under the same condition is what makes a dosage-aware model trainable rather than sign-blind.                                                                      | journal SI (unconfirmed)                           |
| 110 | <b>Jaffe 2019 (multiplexed CRISPR interference epistasis)</b><br>Jaffe M, Sherlock G, Levy SF. G3 2019;9:2843-2852.<br><a href="https://doi.org/10.1534/g3.119.400323">doi:10.1534/g3.119.400323</a>                                                                                     | Two guides per cell, so the perturbation space is combinatorial rather than one edit at a time. This is the shape a Perturb-seq input axis should have, measured here against fitness only.                                                                  | G3 open-access SI (unconfirmed)                    |
| 111 | <b>Sadhu 2016 (CRISPR-directed causal variant mapping)</b><br>Sadhu MJ, Bloom JS, Day L, Kruglyak L. Science 2016;352:1113-1116.<br><a href="https://doi.org/10.1126/science.aaf5124">doi:10.1126/science.aaf5124</a>                                                                    | Drives recombination to a chosen point, so a locus can be swapped against a fixed background rather than inherited with everything linked to it. Turns correlational QTL into causal tests.                                                                  | Science SI (unconfirmed)                           |
| 112 | <b>Weinberg 2016 (ribosome profiling, improved)</b><br>Weinberg DE, Shah P, Eichhorn SW, Hussmann JA, Plotkin JB, Bartel DP. Cell Rep 2016;14:1787-1799.<br><a href="https://doi.org/10.1016/j.celrep.2016.01.043">doi:10.1016/j.celrep.2016.01.043</a>                                  | The corrected reference translation-efficiency dataset, after earlier profiling runs were shown to carry a drug-induced artifact. Translation efficiency is the step between the transcriptome and proteome layers this table otherwise joins by assumption. | GEO (unconfirmed)                                  |
| 113 | <b>Ghaemmaghani 2003 (absolute protein abundance)</b><br>Ghaemmaghani S, Huh WK, Bower K, et al., Weissman JS. Nature 2003;425:737-741.<br><a href="https://doi.org/10.1038/nature02046">doi:10.1038/nature02046</a>                                                                     | Converts every relative proteomics row here into molecules per cell. Without an absolute anchor, enzyme-cost and flux-capacity arguments have no units.                                                                                                      | Nature SI (Excel)                                  |
| 114 | <b>Huh 2003 (GFP localization atlas)</b><br>Huh WK, Falvo JV, Gerke LC, et al., O'Shea EK. Nature 2003;425:686-691.<br><a href="https://doi.org/10.1038/nature02026">doi:10.1038/nature02026</a>                                                                                         | The wild-type localization baseline every perturbation-imaging dataset measures change against. Compartment identity is a prerequisite for compartmentalized pathway engineering.                                                                            | SGD /YeastGFP mirror                               |
| 115 | <b>Michaelis 2023 (protein interactome)</b><br>Michaelis AC, et al., Mann M. Nature 2023;624:192-200.<br><a href="https://doi.org/10.1038/s41586-023-06739-5">doi:10.1038/s41586-023-06739-5</a>                                                                                         | A physical-interaction layer to constrain which genetic interactions reflect complex membership. A graph prior rather than a phenotype.                                                                                                                      | yeast-interactome.biochem.mpg.de + ProteomeXchange |

Table 2, continued

| #   | Dataset, citation and link                                                                                                                                                                                                                                          | Why                                                                                                                                                                                                                  | Data                                  |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| 116 | <b>Christiano 2014 (protein turnover)</b><br>Christiano R, Nagaraj N, Frohlich F, Walther TC. Cell Rep 2014;9:1959-1965.<br><a href="https://doi.org/10.1016/j.celrep.2014.10.065">doi:10.1016/j.celrep.2014.10.065</a>                                             | Half-lives genome-wide. Abundance is production over degradation, and every other proteomics row here measures only the product, so turnover is the missing half.                                                    | Cell Reports SI (unconfirmed)         |
| 117 | <b>Boer 2010 (metabolome across nutrient limitations)</b><br>Boer VM, Crutchfield CA, Bradley PH, Botstein D, Rabinowitz JD. Mol Biol Cell 2010;21:198-211.<br><a href="https://doi.org/10.1091/mbc.e09-07-0597">doi:10.1091/mbc.e09-07-0597</a>                    | The metabolite counterpart to Brauer on the same chemostat design, so metabolite pools are separated from growth rate the same way. Covers central-carbon nodes the deletion metabolomics rows measure at low power. | MBoC SI (unconfirmed)                 |
| 118 | <b>Kitzman 2015 (GAL4 DMS)</b><br>Kitzman JO, Starita LM, Lo RS, Fields S, Shendure J. Nat Methods 2015;12:203-206.<br><a href="https://doi.org/10.1038/nmeth.3223">doi:10.1038/nmeth.3223</a>                                                                      | A transcription factor scanned residue by residue, which pairs with the regulatory-DNA rows: one measures the site, the other the protein that reads it.                                                             | Nat Methods SI (unconfirmed)          |
| 119 | <b>Sharon 2012 (designed promoter library)</b><br>Sharon E, Kalma Y, Sharp A, et al., Segal E. Nat Biotechnol 2012;30:521-530.<br><a href="https://doi.org/10.1038/nbt.2205">doi:10.1038/nbt.2205</a>                                                               | Designed rather than random, so site number, affinity, spacing and orientation vary one at a time. Small, but it is the controlled counterpart the random libraries lack.                                            | Nat Biotechnol SI (unconfirmed)       |
| 120 | <b>Tengolics 2024 (domestication metabolome)</b><br>Tengolics R, Szappanos B, Mulleder M, et al., Ralser M, Papp B. PNAS 2024;121:e2313354121.<br><a href="https://doi.org/10.1073/pnas.2313354121">doi:10.1073/pnas.2313354121</a>                                 | Shows which metabolic traits already moved under domestication, which is a prior on which are tractable to engineer further. Small n.                                                                                | PNAS Dataset S11                      |
| 121 | <b>Sadhu 2018 (CRISPR-directed mitotic recombination)</b><br>Sadhu MJ, Bloom JS, Day L, Siegel JJ, Kosuri S, Kruglyak L. eLife 2018;7:e41090.<br><a href="https://doi.org/10.7554/eLife.41090">doi:10.7554/eLife.41090</a>                                          | Uses targeted recombination to build a genotype series with a single locus swapped against a fixed background, isolating causal effects the way a QTL panel cannot.                                                  | eLife data availability (unconfirmed) |
| 122 | <b>Duveau 2021 (TDH3 promoter allele series)</b><br>Duveau F, Vande Zande P, Metzger BP, et al., Wittkopp PJ. eLife 2021;10:e67806.<br><a href="https://doi.org/10.7554/eLife.67806">doi:10.7554/eLife.67806</a>                                                    | One native promoter dissected at its own locus, with expression and fitness on the same alleles across environments. The rare case tying a regulatory change to a fitness consequence rather than a reporter number. | eLife data availability (unconfirmed) |
| 123 | <b>Strope 2015 (100 clinical and natural genomes)</b><br>Strope PK, Skelly DA, Kozmin SG, Mahadevan G, Stone EA, Magwene PM, Dietrich FS, McCusker JH. Genome Res 2015;25:762-774.<br><a href="https://doi.org/10.1101/gr.185538.114">doi:10.1101/gr.185538.114</a> | A sequenced panel enriched for clinical isolates, which sit outside the vineyard and laboratory backgrounds most panels oversample. Diversity of origin is what makes a natural-variation training set generalize.   | Genome Research SI; SRA (unconfirmed) |
| 124 | <b>Schulte 2023 (mitochondrial complexome)</b><br>Schulte U, et al. Nature 2023;614:153-159.<br><a href="https://doi.org/10.1038/s41586-022-05641-w">doi:10.1038/s41586-022-05641-w</a>                                                                             | Mitochondrial compartmentalization is a live strategy for sequestering toxic intermediates, and this maps the import and assembly bottlenecks that would cap it.                                                     | complexomics.org + ProteomeXchange    |
| 125 | <b>Metzger 2015 (promoter mutation effects on expression and noise)</b><br>Metzger BPH, Yuan DC, Gruber JD, Duveau F, Wittkopp PJ. Nature 2015;521:344-346.<br><a href="https://doi.org/10.1038/nature14424">doi:10.1038/nature14424</a>                            | Single mutations in one native promoter, each with expression and noise measured. The mutational spectrum underlying the natural promoter variation the MPRA rows survey.                                            | Nature SI (unconfirmed)               |

Table 2, continued

| #   | Dataset, citation and link                                                                                                                                                                                                                                                               | Why                                                                                                                                                                                                                                                                                                                                                | Data                                         |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| 126 | <b>Venkataram 2016 (barcoded adaptive lineages, sequenced)</b><br>Venkataram S, Dunn B, Li Y, et al., Petrov DA, Kryazhimskiy S, Sherlock G. Cell 2016;166:1585-1596.<br><a href="https://doi.org/10.1016/j.cell.2016.08.002">doi:10.1016/j.cell.2016.08.002</a>                         | Only the sequenced subset qualifies. The half-million barcoded lineages behind it do not, because their mutations were never determined, and that boundary is the sequence gate doing its job.                                                                                                                                                     | Cell SI; SRA (unconfirmed)                   |
| 127 | <b>Jakociunas 2021 (degron-tuned terpene flux)</b><br>Jakociunas T, Klitgaard AK, Kristensen M, Jensen MK, Keasling JD. 2021.<br><a href="https://doi.org/10.1002/bit.27735">doi:10.1002/bit.27735</a>                                                                                   | Erg20p is FPP synthase, so degron-tuning it reports FPP pathway throughput directly. Small, but it is the only row measuring the mevalonate branch, which the precursor accounting lists as unmeasured.                                                                                                                                            | journal SI (unconfirmed)                     |
| 128 | <b>Shen 2016 (SCRaMbLE genome minimization)</b><br>Shen Y, Stracquadanio G, Wang Y, et al., Boeke JD, Bader JS. Genome Res 2016;26:36-49.<br><a href="https://doi.org/10.1101/gr.193433.115">doi:10.1101/gr.193433.115</a>                                                               | Combinatorial deletion and inversion across a synthetic chromosome, with the surviving genotypes sequenced. Maps which gene combinations are jointly dispensable, which single deletions cannot.                                                                                                                                                   | Genome Research SI (unconfirmed)             |
| 129 | <b>Blount 2018 (SCRaMbLE for phenotype improvement)</b><br>Blount BA, Gowers GF, Ho JCH, Ledesma-Amaro R, Jovicevic D, McKiernan RM, Xie ZX, Li BZ, Yuan YJ, Ellis T. Nat Commun 2018;9:1932.<br><a href="https://doi.org/10.1038/s41467-018-03143-w">doi:10.1038/s41467-018-03143-w</a> | SCRaMbLE driven toward an industrial phenotype rather than toward minimization, so the genotype-to-product link is measured on multiply rearranged genomes.                                                                                                                                                                                        | Nat Commun SI (unconfirmed)                  |
| 130 | <b>Eder 2020 (flux QTL)</b><br>Eder M, Nidelet T, Sanchez I, Camarasa C, Legras JL, Dequin S. PMID 32034164.<br><a href="https://pubmed.ncbi.nlm.nih.gov/32034164/">PMID 32034164</a>                                                                                                    | Flux as a QTL phenotype, with PDB1 and VID30 validated. The label is modeled rather than measured, which caps how far it can be trusted.                                                                                                                                                                                                           | Journal SI (venue and accession unconfirmed) |
| 131 | <b>Liti 2009 (population genomics of <i>S. cerevisiae</i> and <i>S. paradoxus</i>)</b><br>Liti G, Carter DM, Moses AM, et al., Louis EJ. Nature 2009;458:337-341.<br><a href="https://doi.org/10.1038/nature07743">doi:10.1038/nature07743</a>                                           | The founding sequenced isolate panel and the source of the SGRP strains many later crosses use as parents. Historical, and it is where several parental genomes in the segregant rows trace to.                                                                                                                                                    | SGRP /ENA (unconfirmed)                      |
| 132 | <b>Yue 2017 (wild vs domesticated assemblies)</b><br>Yue JX, Li J, Aigrain L, et al., Liti G. Nat Genet 2017;49:913-924.<br><a href="https://doi.org/10.1038/ng.3847">doi:10.1038/ng.3847</a>                                                                                            | Structural variation short reads miss. Relevant because industrial strains differ from S288C structurally, not only at SNPs.                                                                                                                                                                                                                       | ENA/GenBank (accession unconfirmed)          |
| 133 | <b>Brettner 2024 (ultra-high-throughput yeast scRNA-seq)</b><br>Brettner L, et al. Yeast 2024. <a href="https://doi.org/10.1002/yea.3927">doi:10.1002/yea.3927</a> .                                                                                                                     | The platform the costing model in experiments/024 is built on. Not a dataset to ingest; the reason it is here is that a yeast Perturb-seq proposal has to name its chemistry.                                                                                                                                                                      | Yeast (Wiley) SI                             |
| 134 | <b>Mulleder 2012 (prototrophic deletion collection)</b><br>Mulleder M, Capuano F, Pir P, et al., Ralser M. Nat Biotechnol 2012;30:1176-1178.<br><a href="https://doi.org/10.1038/nbt.2442">doi:10.1038/nbt.2442</a>                                                                      | The background the already-built Mulleder 2016 and Messner 2023 were measured in, and mulleder2016.py already cites it. Listed not as an ingestion target but because that loader records the prototrophy-restoring markers as unmodeled: they are a GeneAddition this schema does not yet carry, and auxotrophy markers distort metabolite pools. | EUROSCARF; Addgene kit 40276                 |
| 135 | <b>Zackrisson 2016 (Scan-o-matic growth curves)</b><br>Zackrisson M, Hallin J, Ottosson LG, et al., Warringer J, Blomberg A. G3 2016;6:3003-3014.<br><a href="https://doi.org/10.1534/g3.116.032342">doi:10.1534/g3.116.032342</a>                                                       | Time-series rather than endpoint fitness. A platform, so a specific released condition panel has to be chosen before this becomes a row rather than a plan.                                                                                                                                                                                        | Open-source platform; per-study datasets     |

Table 2, continued

| #   | Dataset, citation and link                                                                                                                                                                                                                                           | Why                                                                                                                                                                                                                                                    | Data                                  |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| 136 | <b>Spellman 1998 (cell-cycle transcriptome)</b><br>Spellman PT, Sherlock G, Zhang MQ, Iyer VR, Anders K, Eisen MB, Brown PO, Botstein D, Futcher B. Mol Biol Cell 1998;9:3273-3297.<br><a href="https://doi.org/10.1091/mbc.9.12.3273">doi:10.1091/mbc.9.12.3273</a> | The cell-cycle expression reference. O’Duibhir showed a large share of deletion expression response is a cell-cycle-distribution artifact, and this is the signature used to detect and remove it.                                                     | SGD /SPELL                            |
| 137 | <b>Levy 2008 (morphology and phenotypic capacitance)</b><br>Levy SF, Siegal ML. PLoS Biol 2008;6:e264.<br><a href="https://doi.org/10.1371/journal.pbio.0060264">doi:10.1371/journal.pbio.0060264</a>                                                                | Scores how much a deletion increases cell-to-cell variability rather than shifting the mean, identifying capacitor genes. Variance as the phenotype is a readout only per-cell data supports, which is the argument the Perturb-seq proposal rests on. | PLoS Biology SI (unconfirmed)         |
| 138 | <b>Causton 2001 (environmental change transcriptome)</b><br>Causton HC, Ren B, Koh SS, et al., Young RA. Mol Biol Cell 2001;12:323-337.<br><a href="https://doi.org/10.1091/mbc.12.2.323">doi:10.1091/mbc.12.2.323</a>                                               | An independently generated environmental-response compendium contemporaneous with Gasch 2000. Two independent measurements of the same response class is a platform-effect control.                                                                    | SGD /SPELL                            |
| 139 | <b>Gasch 2001 (DNA damage response transcriptome)</b><br>Gasch AP, Huang M, Metzner S, Botstein D, Elledge SJ, Brown PO. Mol Biol Cell 2001;12:2987-3003.<br><a href="https://doi.org/10.1091/mbc.12.10.2987">doi:10.1091/mbc.12.10.2987</a>                         | Extends the stress-response backbone to genotoxic stress, and includes checkpoint mutants, so it carries a small genotype axis rather than being wild-type only.                                                                                       | SGD /SPELL                            |
| 140 | <b>Handfield 2013 (image-based localization change)</b><br>Handfield LF, Chong YT, Simmons J, Andrews BJ, Moses AM. PLoS Comput Biol 2013;9:e1003085.<br><a href="https://doi.org/10.1371/journal.pcbi.1003085">doi:10.1371/journal.pcbi.1003085</a>                 | An unsupervised treatment of the same imaging collection, which matters because the supervised class labels in CYCLOPs constrain what can be found.                                                                                                    | PLoS Comput Biol SI (unconfirmed)     |
| 141 | <b>Kaplan 2009 (nucleosome sequence preferences)</b><br>Kaplan N, Moore IK, Fondufe-Mittendorf Y, et al., Segal E. Nature 2009;458:362-366.<br><a href="https://doi.org/10.1038/nature07667">doi:10.1038/nature07667</a>                                             | Separates the sequence-intrinsic component of nucleosome positioning from the cellular one by reconstituting in vitro. That separation is what lets a promoter model attribute an effect to sequence rather than to context.                           | GEO (unconfirmed)                     |
| 142 | <b>Giaever 2004 (chemogenomic drug-target profiling)</b><br>Giaever G, Flaherty P, Kumm J, et al., Davis RW. PNAS 2004;101:793-798.<br><a href="https://doi.org/10.1073/pnas.0307490100">doi:10.1073/pnas.0307490100</a>                                             | An early HIP profiling set from the group that built the collection. Superseded in scale by Lee and Hoepfner, retained as a low-cost cross-validation panel.                                                                                           | PNAS SI (unconfirmed)                 |
| 143 | <b>Yu 2021 (proteome and metabolome under nitrogen limitation)</b><br>Yu R, Vorontsov E, Sihlbom C, Nielsen J. eLife 2021;10:e65722.<br><a href="https://doi.org/10.7554/eLife.65722">doi:10.7554/eLife.65722</a>                                                    | Nitrogen limitation is the standard trigger for redirecting carbon to product, and this measures both layers under it. Condition axis only, and small.                                                                                                 | eLife data availability (unconfirmed) |
| 144 | <b>Paulo 2016 (TMT proteome across carbon sources)</b><br>Paulo JA, O’Connell JD, Gaun A, Gygi SP. Mol Biol Cell 2016;27:2823-2838.<br><a href="https://doi.org/10.1091/mbc.E16-03-0184">doi:10.1091/mbc.E16-03-0184</a>                                             | Proteome remodeling across carbon sources, which is the substrate switch every bioprocess makes. Condition axis only.                                                                                                                                  | ProteomeXchange (unconfirmed)         |
| 145 | <b>Gibney 2013 (osmotic and glucose stress fitness)</b><br>Gibney PA, Lu C, Caudy AA, Hess DC, Botstein D. PNAS 2013;110:E4393-E4402.<br><a href="https://doi.org/10.1073/pnas.1318100110">doi:10.1073/pnas.1318100110</a>                                           | High-gravity fermentation is an osmotic problem, and this covers the osmotic axis on the deletion collection with a growth readout rather than a transcriptome.                                                                                        | PNAS SI (unconfirmed)                 |

Table 2, continued

| #   | Dataset, citation and link                                                                                                                                                                                                                                                                                                                           | Why                                                                                                                                                                                         | Data                                                      |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| 146 | <b>Meurer 2018 (genome-wide GFP library, rebuilt)</b><br>Meurer M, Duan Y, Sass E, Kats I, Herbst K, Buchmuller BC, Dederer V, Huber F, Kirrmaier D, Steff M, Van Laer K, Dick TP, Lemberg MK, Khmelinskii A, Levy ED, Knop M. Nat Methods 2018;15:617-620.<br><a href="https://doi.org/10.1038/s41592-018-0044-9">doi:10.1038/s41592-018-0044-9</a> | A rebuilt and sequence-verified tagging collection, correcting errors accumulated in the original. Verify the DOI before ingestion; it collides with the Weill 2018 SWAT row in this table. | Nat Methods SI (DOI collision with Weill 2018 UNRESOLVED) |
| 147 | <b>Bennett 2001 (ionizing-radiation resistance)</b><br>Bennett CB, et al. Nat Genet 2001;29:426-434.<br><a href="https://doi.org/10.1038/ng778">doi:10.1038/ng778</a>                                                                                                                                                                                | Genome-stability tolerance. Only indirectly relevant, through strain robustness over long fed-batch runs where instability erodes titer.                                                    | Nat Genet SI (likely PDF)                                 |
| 148 | <b>Sugiyama (lactic-acid tolerance)</b><br>Sugiyama M, Kaneko Y, et al. National Research Institute of Brewing technical report.<br><a href="https://nissr.or.jp/wp-content/uploads/NISR06sugiyama.pdf">nissr.or.jp/wp-content/uploads/NISR06sugiyama.pdf</a>                                                                                        | Lactic acid is a major platform chemical, but this is a non-peer-reviewed technical report with PDF-only tables. Lowest provenance confidence in the table; treat as a hypothesis source.   | NRIB technical report PDF                                 |
| 149 | <b>Sousa 2013 (acetic-acid programmed cell death screen)</b><br>Sousa M, Duarte AM, Fernandes TR, et al. 2013. PMID 24286259.<br><a href="https://pubmed.ncbi.nlm.nih.gov/24286259/">PMID 24286259</a>                                                                                                                                               | Survival rather than growth rate under the same stressor. A different phenotype from fitness, and the one that matters when a fed-batch culture crashes rather than slows.                  | journal SI (unconfirmed)                                  |
| 150 | <b>Nagalakshmi 2008 (RNA-seq transcriptome annotation)</b><br>Nagalakshmi U, Wang Z, Waern K, Shou C, Raha D, Gerstein M, Snyder M. Science 2008;320:1344-1349.<br><a href="https://doi.org/10.1126/science.1158441">doi:10.1126/science.1158441</a>                                                                                                 | The first RNA-seq transcriptome, and the annotation correction underneath the microarray-era expression rows. Historical, and it is why gene models used by later datasets differ.          | SRA /GEO (unconfirmed)                                    |
| 151 | <b>Yofe 2016 (SWAT N-terminal library)</b><br>Yofe I, Weill U, Meurer M, et al., Knop M, Schuldiner M. Nat Methods 2016;13:371-378.<br><a href="https://doi.org/10.1038/nmeth.3795">doi:10.1038/nmeth.3795</a>                                                                                                                                       | N-terminal tagging, which preserves C-terminal signals the GFP collection destroys. Complements Huh 2003 on exactly the proteins that collection mislocalizes.                              | Nat Methods SI; EUROSCARF (unconfirmed)                   |
| 152 | <b>Hénault 2023 (hybrid and introgression panel)</b><br>Henault M, Marsit S, Charron G, Landry CR. Nat Commun / Landry lab hybrid panels.<br><a href="https://doi.org/10.1038/s41467-024-46155-5">doi:10.1038/s41467-024-46155-5</a>                                                                                                                 | Hybrid genomes rather than recombinants of one cross, so the genotype axis includes whole-subgenome combinations. Confirm the exact release before ingestion.                               | journal SI (unconfirmed)                                  |
| 153 | <b>Cubillos 2013 (F12 recombinant panel construction)</b><br>Cubillos FA, Parts L, Salinas F, et al., Liti G. Genetics 2013;195:1141-1155.<br><a href="https://doi.org/10.1534/genetics.113.155515">doi:10.1534/genetics.113.155515</a>                                                                                                              | The panel and genotype table the already-listed Cubillos 2017 nitrogen study resolves against. Ingesting it is what makes that row's genotype axis self-contained.                          | Genetics SI Table S1                                      |
| 154 | <b>Kessi-Perez 2019 (nitrogen-consumption QTL panel)</b><br>Kessi-Perez EI, Molinet J, Martinez C, et al. 2019/2020 nitrogen-utilization QTL work.<br><a href="https://doi.org/10.1186/s13068-019-1440-9">doi:10.1186/s13068-019-1440-9</a>                                                                                                          | A second nitrogen-utilization mapping panel alongside Cubillos 2017. Nitrogen sets higher-alcohol and ester formation, so it is fermentation-relevant, and the exact release needs pinning. | journal SI (unconfirmed)                                  |
| 155 | <b>Lian 2016 (combinatorial xylose pathway optimization)</b><br>Lian J, Chao R, Zhao H. Metab Eng 2016;33:139-148.<br><a href="https://doi.org/10.1016/j.ymben.2015.11.001">doi:10.1016/j.ymben.2015.11.001</a>                                                                                                                                      | Combinatorial promoter and copy-number variants of one pathway with titers measured per combination. Small, and it is the record type inverse design produces.                              | Metab Eng SI (unconfirmed)                                |



### 3 What the recommended 151 add up to ✗

**Table 3:** Candidates by class, split at the cut line. *Genotypes* and *Instances* sum the per-row axes; rows with no count contribute nothing, so both totals are lower bounds.

| Class                    | In top 151 | Reserve | Genotypes   | Instances         |
|--------------------------|------------|---------|-------------|-------------------|
| Natural variation        | 31         | 3       | 1,532,358   | $5.0 \times 10^6$ |
| Tolerance / robustness   | 24         | 0       | 128,063     | $5.6 \times 10^7$ |
| CRISPR library screen    | 11         | 0       | 328,794     | $3.9 \times 10^5$ |
| Expression / single cell | 15         | 0       | 5,411       | $6.1 \times 10^4$ |
| Metabolite / precursor   | 14         | 1       | 45,331      | $1.4 \times 10^5$ |
| Modality / backbone      | 36         | 0       | 789,981     | $1.4 \times 10^6$ |
| Total                    | 151        | 4       | 124,151,155 | $1.8 \times 10^8$ |

The recommended set carries roughly  $1.8 \times 10^8$  genotype-by-environment instances and  $5.7 \times 10^8$  individual measurements. Both are lower bounds: rows whose axes could not be pinned down contribute nothing to either total, and the instance figures marked † are products of the two axes rather than counts the papers report.

The totals are dominated by a handful of rows, so quoting them without naming those rows would mislead. de Boer 2020 alone supplies  $10^8$  instances, Turco 2023  $3.8 \times 10^7$  and Lee 2014  $1.3 \times 10^7$ . Of those three, Turco’s contribution is not net new, since it is a meta-aggregation of screens this table lists separately, and Lee’s is not currently reachable (Sec. 6.1). de Boer’s stands, and it is a different kind of instance from the rest: an 80-base promoter with an expression readout, not a strain with a phenotype.

The honest summary is that this list contains one dataset larger than everything already built, one larger than everything else here, and about  $2 \times 10^7$  instances spread across the remaining 148 rows.

### 4 The classes ✗

Six classes were requested. Three more were added because the expansion turned up coherent groups that did not fit any of them: regulatory DNA, deep mutational scans and combinatorial genomes, all covered in Sec. 5.2.

#### 4.1 Natural isolates, and where sequence diversity crosses over ✗

Natural variation is the largest class after the backbone rows (Table 3), and the crossover the class was asked for is concentrated in three of them.

Dutta 2026 barcodes 520 natural isolates and profiles them against more than 600 compounds. Those isolates are drawn from the sequenced 1,011 panel, which is the same panel behind the already-built Caudal 2024 pan-transcriptome and behind Muenzner 2024. One genotype axis therefore carries a genome, a transcriptome, a proteome, and now a chemogenomic response surface, which is the join that makes a transfer question askable at all.

Hale 2024 is the sharpest instrument for that question. It crosses 8,046 CRISPRi guides over 1,721 genes into 169 sequenced segregants, so it measures directly how much a perturbation’s effect depends on the background it is made in. Galardini 2019 asks the same question with deletions instead of knockdowns, building 3,786 knockouts independently in four sequenced backgrounds across 38 conditions, and reports that about 18.5 percent of deletion phenotypes change between backgrounds.

That 18.5 percent is the reason this class matters beyond breadth. Nearly every fitness and metabolite dataset in the built set is S288C, and a model trained only there has no way to know which of its predictions survive a change of chassis.

#### 4.2 Tolerance on recalcitrant biomass, and product tolerance ✗

Lignocellulosic inhibitor coverage in the built set currently runs through broad compound panels rather than dedicated screens, so three rows close named gaps: Pereira 2014 screens the deletion collection against real industrial wheat-straw hydrolysate rather than reconstituted single toxins, Endo 2008 covers vanillin, and Xiao 2014 covers furfural through a knockdown axis that reaches essential genes a deletion collection cannot.



Product tolerance is answered mainly on isobutanol, and the earlier assumption that no genome-wide isobutanol screen exists is wrong. Kuroda 2019 screens the deletion collection for isobutanol-specific fitness against an ethanol comparator, so the alcohol-general stress response subtracts out; its *GLN3* hit raised production 4.9-fold, which is a rare case of a tolerance screen translating into a titer. Liu 2021 is an independent full-collection screen of the same phenotype with a different readout, giving a same-genotype, same-phenotype, different-method replicate. Crook 2016 adds a dose-graded knockdown axis over isobutanol and 1-butanol together, and Peeters 2021 supplies isobutanol titer as a mapped trait across 1,125 sequenced segregants.

Van Leeuwen 2024 runs acetic, butyric and octanoic acid on one strain panel, which is what lets shared and acid-specific mechanisms separate inside a single study instead of across screens that cannot be compared.

**Acetic acid, production as well as tolerance.** Acetic acid is now covered on both sides, which is unusual. On tolerance: Mira 2010 screens the full deletion collection at controlled pH and reports about 650 determinants, Mukherjee 2021 covers the essential genes by CRISPRi, Sousa 2013 covers gain-of-function by overexpression, and the already-built Mormino 2022 covers it by biosensor. Four perturbation types, one phenotype.

Production is thinner but real. Peeters 2021 measures acetic acid titer as one of its 18 mapped traits across 1,125 sequenced segregants, which is the only quantitative genotype-to-acetate-titer map at scale. Chang 2012 screens the whole deletion collection for altered organic-acid output including acetate, though the readout is an ordinal halo rather than a titer. No genome-wide screen was found whose primary label is a quantitative acetate titer per strain, so the pairing that exists is Peeters for production against the four tolerance screens, not a matched production screen against them.

### 4.3 CRISPR library screens ✗

The already-built Lian 2019 MAGIC screen reads out furfural tolerance by growth selection; Dong 2021 is the sibling that swaps the growth selection for an S-adenosyl-methionine biosensor and sorts by fluorescence, so the label becomes product concentration rather than fitness. That is the pattern worth copying for every precursor in Sec. 4.5, and it is the biosensor MAGIC screen the scoping request asked after.

Bao 2018 and Roy 2018 both move the genotype axis below the ORF, to designed single-nucleotide and small-indel variants. Both are designed rather than per-strain-verified, so both would enter as designed genotypes with a verification status. McGlincy 2021 and Momen-Roknabadi 2020 are two independently designed CRISPRi libraries; ingesting both makes library-to-library transfer a cheap and decisive test of whether a model has learned gene function or guide-design idiosyncrasy.

### 4.4 Expression across conditions ✗

This class is genuinely thin in a way worth stating rather than papering over. Nadal-Ribelles 2025 is the largest genotype-indexed expression dataset in yeast at roughly 3,500 knockouts under osmotic stress, and it is already built. The candidates here supply what it lacks rather than more of it: Gasch 2000 is the canonical wild-type stress-response baseline that any genotype-indexed transcriptome has to be read against, N'Guessan 2025 profiles roughly 4,500 sequenced segregants by single-cell RNA-seq, and Leutert 2023 covers 101 conditions at the phosphosite level, where enzyme regulation acts faster than transcription.

The initial reading was that datasets in the microbe-perturb-seq collection carry little data over large genotype numbers. That holds. Nothing found in this pass changes it, and the gap it points to is addressed in Sec. 5 rather than by another row.

### 4.5 Metabolite and precursor coverage ✗

The precursor accounting in `notes/metabolism.central-carbon-precursors.md` sets the target: of the twelve central carbon precursors, acetyl-CoA is measured on 19 strains and succinyl-CoA and malonyl-CoA on none, so identity coverage is good and statistical power is not.

Blank 2005 is the only dataset in this table that measures flux rather than a concentration or a titer, across roughly 200 deletion mutants, and it is the only real handle on acetyl-CoA turnover under genetic perturbation. Malonyl-CoA still has no direct measurement anywhere; the nearest handles remain its dominant sinks, which the already-built Xue 2025 free fatty acids and da Silveira 2014 lipidomics supply, and Zhu 2014 extends with a lipidome across 129 kinase and phosphatase deletions.

Cooper 2010 is the highest-value row in the class for a reason unrelated to novelty. It measures the same amino-acid pools as the already-built Mulleder 2016, on an overlapping genotype axis, by capillary electrophoresis instead of mass spectrometry. Two independent platforms measuring one trait class is the cleanest available test of whether a model has learned biology or a batch effect, and nothing else in the built set offers it.

On carotenoids, Ozaydin 2013 is built. Triikka 2015 is the other high-throughput carotenoid screen: 4,700 heterozygous deletions scored by color, then carried through iterative rounds to a sclareol titer. Halving gene dosage rather than deleting outright is what finds the isoprenoid flux control points a null would kill, and no other row in this table uses that design. It is currently recorded as figure-only, so confirming that Additional file 1 carries per-strain scores is a precondition, not a formality.

## 4.6 Regulatory DNA, deep scans and combinatorial genomes ✗

Three new classes, all with exactly known sequences and all going deep on a locus rather than shallow across a genome. They are covered in Sec. 5.2, because their value is best stated against the Perturb-seq specification rather than as another phenotype class.

## 4.7 Perturbation modality and reference backbone ✗

None of these is a headline dataset and one is load-bearing. Puddu 2019 is whole-genome sequence for every strain in the deletion collection. Twenty-one recommended rows declare an **S288C-KO** sequence basis, and without Puddu that basis is an assumption that each strain differs from the reference at exactly the intended locus. Puddu shows it frequently does not. Ingesting it converts a shared assumption across half this table into a measurement, which is worth more than its own 4,800 instances suggest.

# 5 Rows that bear on a yeast Perturb-seq ✗

A Perturb-seq is defined by two dimensionalities, and a dataset earns a place here only by being high on at least one of them.

- **Input dimensionality** is the size of the perturbation space per cell: combinatorial edits, or a perturbation crossed with genetic background. A large library of single perturbations is not high-dimensional input. It is one edit per cell, sampled widely, and a model fit to it never sees two edits interact.
- **Output dimensionality** is the readout: a transcriptome or a per-cell distribution rather than a scalar fitness or titer.

That distinction removes a row this table previously carried. Momen-Roknabadi 2020 is a genome-wide CRISPRi library, which is large, but it delivers one guide per cell and reads out competitive fitness. It is low-dimensional on both axes and is no longer flagged; its value is as a second library for testing guide-design transfer, not as Perturb-seq evidence.

**Table 4:** Rows bearing on a yeast Perturb-seq, grouped by axis rather than by rank. *Rank* is the row’s position in Table 1. A large library of single perturbations read out by fitness is not high-dimensional input: it is one edit per cell, sampled widely.

| Rank                                                                                                         | Dataset                                          | What it supplies                                                                                                                                                                                                                                                     |
|--------------------------------------------------------------------------------------------------------------|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>High on BOTH axes: combinatorial or background-crossed perturbation AND a transcriptome-scale readout</b> |                                                  |                                                                                                                                                                                                                                                                      |
| 1                                                                                                            | <b>Boocock 2025 (single-cell eQTL mapping)</b>   | 27,744 single cells drawn from 393 sequenced recombinants in two carbon sources, a median of 17 cells per segregant. High-dimensional on both axes at once, which almost nothing in yeast is, and the closest existing template for the proposed design.             |
| 5                                                                                                            | <b>Albert 2018 (eQTL in 1,012 segregants)</b>    | Transcriptomes on a thousand sequenced recombinants, so genotype maps to a genome-wide expression vector rather than one trait. The single best existing analog to a Perturb-seq readout on a high-dimensional genotype axis.                                        |
| 29                                                                                                           | <b>N’Guessan 2025 (segregant scRNA-seq eQTL)</b> | Single-cell transcriptomes indexed by recombinant natural genotype rather than by an engineered knockout. The closest existing analog to the proposed Perturb-seq design, and the only one that already pairs per-cell expression with sequenced genotypes at scale. |
| <b>High-dimensional perturbation space, scalar readout</b>                                                   |                                                  |                                                                                                                                                                                                                                                                      |

| Rank                                                                         | Dataset                                                | What it supplies                                                                                                                                                                                                                                                                                  |
|------------------------------------------------------------------------------|--------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2                                                                            | de Boer 2020 (100M random promoters)                   | The largest sequence-to-phenotype dataset in yeast by orders of magnitude, and every genotype is an exactly known 80mer. The perturbation space is continuous rather than a gene list, which is what a model of regulatory grammar needs and no deletion collection can supply.                   |
| 3                                                                            | Vaishnav 2022 (regulatory DNA fitness landscape)       | Extends de Boer from measurement to a fitness landscape, and tests native and natural-isolate promoter variants against the random library. The bridge between synthetic sequence space and the natural variation the isolate panels carry.                                                       |
| 8                                                                            | Hale 2024 (CRISPRi x natural variation)                | Directly measures how a genetic perturbation's effect changes with genetic background. The single best dataset for asking whether a model trained on one background transfers to another.                                                                                                         |
| 10                                                                           | Nguyen Ba 2022 (barcoded bulk QTL, 100k segregants)    | The largest genotyped recombinant panel in yeast, each segregant barcoded and low-coverage sequenced against known parents. Two orders of magnitude more genotypes than Bloom 2013 on the same kind of cross, which is what makes small-effect and epistatic loci detectable.                     |
| 16                                                                           | Li 2016 (tRNA fitness landscape)                       | A near-complete fitness landscape of one native yeast gene. Every genotype is a known sequence and the sampling is exhaustive rather than one allele per gene, which is the opposite of what a deletion collection provides.                                                                      |
| 19                                                                           | Domingo 2018 (tRNA double-mutant landscape)            | Pairwise epistasis at nucleotide resolution inside one gene. The higher-order structure Costanzo and Kuzmin map between genes, measured within a gene, which is the resolution inverse design actually needs.                                                                                     |
| 33                                                                           | Bloom 2025 (global epistasis in a yeast cross)         | Reports that many natural variants have effects that scale with background fitness, which is a specific, testable claim about how genotype effects compose. Directly relevant to whether a model trained on one background transfers.                                                             |
| 38                                                                           | Faure 2022 (doubledeepPCA)                             | Separates a variant's effect on folding from its effect on binding by measuring both in the same library. That decomposition is what turns a fitness landscape into a mechanistic one.                                                                                                            |
| 42                                                                           | Diss 2018 (Fos-Jun interaction double mutants)         | Exhaustive pairwise mutation of an interaction interface, read out in yeast. The genotype is a heterologous domain rather than a yeast gene, so it enters as an engineered chassis, but the epistasis structure is the deepest in the table.                                                      |
| 48                                                                           | McGlinchy 2021 (genome-scale CRISPRi library)          | Ten guides per gene gives a graded knockdown axis instead of a binary null, and it covers essential genes. This is the library a yeast Perturb-seq would most plausibly be built on.                                                                                                              |
| 50                                                                           | Puchta 2016 (snoRNA U3 landscape)                      | The companion landscape on a non-coding RNA, published alongside Li 2016. Two exhaustive landscapes on different molecule classes is a stronger test of a sequence model than either alone.                                                                                                       |
| 76                                                                           | Renganaath 2020 (natural promoter-variant MPRA)        | Native promoter variation measured allele by allele rather than sampled randomly, with 451 variants called causal. The row that joins the synthetic MPRA libraries to the natural-isolate panels: same readout, real alleles.                                                                     |
| 84                                                                           | Zhang 2022 (GCE-SCRaMbLE recombination outcomes)       | Each cell carries tens of simultaneous structural edits, sequenced afterwards. The deepest per-cell perturbation in the table: a genotype here is a rearranged chromosome, not an allele, which is what inverse strain design eventually has to predict on.                                       |
| 99                                                                           | Guo 2018 (CRISPR-Cas9 tiling of essential genes)       | Tiles guides across genes rather than sampling a few per gene, so the readout resolves within-gene functional regions. A finer input axis than a gene-level library.                                                                                                                              |
| 110                                                                          | Jaffe 2019 (multiplexed CRISPR interference epistasis) | Two guides per cell, so the perturbation space is combinatorial rather than one edit at a time. This is the shape a Perturb-seq input axis should have, measured here against fitness only.                                                                                                       |
| <b>Transcriptome-scale or per-cell readout, low-dimensional perturbation</b> |                                                        |                                                                                                                                                                                                                                                                                                   |
| 9                                                                            | Jackson 2020 (TF-deletion single-cell atlas)           | Single-cell transcriptomes across a panel of regulator deletions in two media, used to learn a regulatory network. Few genotypes, but it is one of the only yeast datasets crossing a genetic perturbation with a per-cell transcriptome, which is the quadrant the Perturb-seq proposal targets. |
| 32                                                                           | Hu 2007 (TF deletion expression compendium)            | Transcriptomes for the regulators specifically, predating and complementing the already-built Kemmeren compendium. Deleting a transcription factor is the perturbation whose transcriptome response is most interpretable.                                                                        |
| 39                                                                           | Chong 2015 / CYCLOPs (single-cell proteome dynamics)   | Per-cell rather than per-strain-mean labels, over 20 million cells. The distributional readout a Perturb-seq design has to be scored against.                                                                                                                                                     |
| 67                                                                           | Newman 2006 (protein noise, GFP library)               | Measures the distribution rather than the mean, per protein, in two media. Noise is the quantity a per-cell readout recovers and a bulk one destroys, so this sets what a single-cell design should expect to see.                                                                                |

| Rank | Dataset                                                         | What it supplies                                                                                                                                                                                                                                       |
|------|-----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 87   | <b>Smith 2008 (gene-environment interaction eQTL)</b>           | Transcriptomes on the same segregants in two carbon sources, so an expression QTL can be shown to be condition-dependent. Small, and it is the design Boocock 2025 scales up.                                                                          |
| 88   | <b>Lenstra 2011 (chromatin regulator deletion expression)</b>   | Transcriptomes for the chromatin machinery specifically, from the group behind the built Kemmeren compendium and in the same format. Chromatin regulators are the class whose deletion most reshapes the expression landscape.                         |
| 90   | <b>Breker 2013 (GFP library under stress)</b>                   | The GFP collection re-imaged under stress rather than at baseline, so relocalization is measured as a response. Feeds LoQAtE, and adds the condition axis Huh 2003 lacks.                                                                              |
| 91   | <b>Tkach 2012 (protein relocalization under DNA damage)</b>     | The GFP collection imaged under damage, giving a second condition axis on the imaging layer alongside Breker’s stress panel. Relocalization is a response no abundance measurement captures.                                                           |
| 94   | <b>Denervaud 2013 (microfluidic GFP dynamics)</b>               | Single-cell protein dynamics under controlled environmental switches, rather than endpoint snapshots. The temporal axis every other imaging row lacks.                                                                                                 |
| 100  | <b>Yvert 2013 (single-cell trait variation across isolates)</b> | CalMorph morphology on natural isolates rather than deletion strains, in the same feature space as the already-built Ohya and Ohnuki datasets. Joins the morphology layer to the natural-variation axis without harmonization.                         |
| 104  | <b>Nadal-Ribelles 2019 (sensitive yeast scRNA-seq)</b>          | Establishes within-clone transcript correlation, which is the noise floor any per-strain Perturb-seq mean has to clear. Directly sizes the cells-per-guide question.                                                                                   |
| 105  | <b>Gasch 2017 (single-cell stress heterogeneity)</b>            | Separates intrinsic from extrinsic expression heterogeneity under stress. Sets what fraction of Perturb-seq variance is recoverable signal rather than cell-state noise.                                                                               |
| 133  | <b>Brettner 2024 (ultra-high-throughput yeast scRNA-seq)</b>    | The platform the costing model in experiments/024 is built on. Not a dataset to ingest; the reason it is here is that a yeast Perturb-seq proposal has to name its chemistry.                                                                          |
| 137  | <b>Levy 2008 (morphology and phenotypic capacitance)</b>        | Scores how much a deletion increases cell-to-cell variability rather than shifting the mean, identifying capacitor genes. Variance as the phenotype is a readout only per-cell data supports, which is the argument the Perturb-seq proposal rests on. |

## 5.1 The quadrant that matters is nearly empty ✗

Of 156 candidates, 15 are high on the input axis, 15 on the output axis, and **three** on both: Boocock 2025, Albert 2018 and N’Guessan 2025. All three reach the both-axes quadrant the same way, by crossing recombinant natural genotypes with a transcriptome, and none of them perturbs a gene deliberately.

So the gap is specific rather than general. Yeast has deep genotype-to-fitness data and deep wild-type transcriptome data, and where the two meet it is through natural variation, not through designed perturbation. The largest genotype-indexed transcriptome with designed perturbations is Kemmeren 2014, already built, and it is 1,484 single deletions read out one gene at a time, which is exactly the shallow case: one knockout, one profile, no interaction measured. Sameith 2015, also built, adds double mutants but only 72 of them.

That is the argument for running the experiment, and it is also the specification. To land in the empty quadrant a design needs a combinatorial perturbation per cell, not a larger single-perturbation library, and a transcriptome readout rather than a growth score.

## 5.2 Depth rather than breadth ✗

Three classes were added to this pass specifically because they go deep on one locus instead of shallow across many, and every one of them carries an exactly known sequence per genotype.

**Deep mutational scans, 9 rows.** Li 2016 and Puchta 2016 are near-complete fitness landscapes of a single native yeast gene at roughly 65,000 and 60,000 variants. Domingo 2018 and Diss 2018 go further and measure pairwise mutations, so epistasis is resolved inside a gene rather than between genes. Faure 2022 separates a variant’s effect on folding from its effect on binding by measuring both in the same library, which is what turns a landscape mechanistic. These supply the input dimensionality the fitness screens lack.

**Regulatory DNA, 8 rows.** de Boer 2020 is above 100 million exactly known promoter sequences with a measured expression level each, which is the largest sequence-to-phenotype dataset in yeast by orders of magnitude. Vaishnav 2022 extends it to a fitness landscape and tests natural-isolate promoter variants against the random library, and Renganaath 2020 measures 5,832 real promoter alleles from natural variation. This is the answer to whether promoter strength has been measured at scale: it has, both for random sequence and for native alleles, and the two are on the same readout.

**Combinatorial genomes, 3 rows.** SCRaMbLE genotypes carry tens of simultaneous structural edits and are sequenced afterwards. A genotype there is a rearranged chromosome rather than an allele, which is the deepest per-cell perturbation available and the hardest thing a genotype-to-phenotype model will eventually be asked to predict on.

### 5.3 The synergy question is answerable, and not yet answered ✗

Whether one dataset predicts another decides how much new data is worth collecting, and the list contains the pairs to test it before any is collected:

- **Same phenotype, different platform.** Cooper 2010 against the built Mulleder 2016, amino acids by capillary electrophoresis against mass spectrometry on an overlapping genotype axis.
- **Same phenotype, different method.** Liu 2021 against Kuroda 2019, isobutanol tolerance on the same collection by colony imaging against competitive fitness.
- **Same perturbation, different background.** Hale 2024, Galardini 2019 and She 2018, all of which vary the background while holding the perturbation fixed.
- **Different library, same modality.** McGlinchey 2021 against Momen-Roknabadi 2020, two CRISPRi libraries with different vectors and design rules.
- **Cheap layer against expensive layer.** Caudal 2024 transcriptomes against Muenzner 2024 proteomes on overlapping isolates, and Skelly 2013, which measures both on the same 22 strains rather than joining across studies.
- **Synthetic sequence against natural allele.** de Boer 2020 and Vaishnav 2022 against Renganaath 2020: whether a model trained on random promoters predicts the effect of a real one.

None of these transfers has been measured. Each is a hypothesis about what a model could learn, and each becomes a run once both halves are ingested. The last pair is the cheapest and the most immediately decisive.

## 6 What was dropped ✗

**Table 5:** Considered and dropped. *no-sequence* is the hard gate: without a route to the strain’s genomic content there is no genotype to map a phenotype from.

| Dataset or group                                                                               | Rule          | Reason                                                                                                                                                                                                                                                                                                                                                     |
|------------------------------------------------------------------------------------------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Snoek 2015 robot-assisted genome shuffling (ethanol tolerance)                                 | no-sequence   | Shuffled progeny are recombinants of unrecorded parental segments with no per-strain sequencing, so the genotype cannot be reconstructed.                                                                                                                                                                                                                  |
| Adaptive laboratory evolution / EMS mutagenesis tolerance panels (e.g. farnesol-FPP selection) | no-sequence   | Mutations are unmapped, so a strain’s genome is unknown even in principle without whole-genome resequencing the study did not do.                                                                                                                                                                                                                          |
| Avalos lab isobutanol-biosensor deletion screen                                                | not-a-dataset | Described only in a PhD thesis and a DOE report. Track for publication; not citable as a dataset.                                                                                                                                                                                                                                                          |
| Kitamoto/Kaneko ester and higher-alcohol screen                                                | not-a-dataset | Citation could not be pinned to a specific paper. A placeholder, not a located dataset.                                                                                                                                                                                                                                                                    |
| Fuhrer/Zamboni flow-injection metabolomics                                                     | off-species   | The validated genome-scale application is in <i>E. coli</i> . No yeast deletion-collection version was located.                                                                                                                                                                                                                                            |
| Anglada-Girotto 2022 (CRISPRi + metabolomics)                                                  | off-species   | 352 CRISPRi genes against 1,342 drug-induced metabolic changes, but in <i>E. coli</i> . Verified from the mirrored PDF.                                                                                                                                                                                                                                    |
| Wildenhain 2016 (Sci Data chemical-genetic matrix)                                             | already-built | A data descriptor, not a second experiment. Its own Data Citation 1 is NCBI PubChem BioAssay AID 1159580, which is exactly what <code>torchcell/datasets/scerevisiae/wildenhain2015.py</code> already ingests: 242 strains, 5,518 compounds, 492,126 interaction tests.                                                                                    |
| O’Duibhir 2014 expression compendium                                                           | already-built | The paper’s expression data IS the Kemmeren 2014 compendium plus PCA transforms, already built, as <code>oduibhir2014.py</code> states; only its growth-rate readout was new and that is built too. Caught by <code>check_candidate_overlap.py</code> rather than by inspection, which is the first time the script found a duplicate before a person did. |
| Ozturk 2022 proteome-effects screen                                                            | off-species   | 3,308 deletions in <i>S. pombe</i> . Out of species scope.                                                                                                                                                                                                                                                                                                 |
| Chen 2024 <i>K. phaffii</i> CRISPR fitness screen; Y1000+ multi-species panel                  | off-species   | Not <i>S. cerevisiae</i> . Revisit if the generalization axis widens to other hosts.                                                                                                                                                                                                                                                                       |

| Dataset or group                                                                                                                               | Rule          | Reason                                                                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| iIsor850 genome-scale model (Issatchenkia orientalis)                                                                                          | not-a-dataset | A reconstruction, not an experiment, and off species. No genome-wide I. orientalis phenotype screen was located; the CABBI knockout library is unpublished.                         |
| CABBI 13C-MFA kinetic-model repository                                                                                                         | not-a-dataset | 16 knockout strains total, and the released fluxes are K-FIT model fits recapitulating 75-77 percent of measurements, not raw measurement. Blank 2005 supersedes it on both counts. |
| Sc2.0 / Sc3.0 synthetic-genome design                                                                                                          | not-a-dataset | Genome-engineering programs, not gene-indexed perturbation-phenotype data.                                                                                                          |
| Lee 2014 / Hoepfner 2014 / Vanacloig-Pedros 2022 / Messner 2023 / Mulleder 2016 / Lian 2019 / Mormino 2022 / Nadal-Ribelles 2025 and 12 others | already-built | Named as top candidates by the 2026-07 triage pass and BUILT since. Only Lee 2014 remains outstanding and is kept as row 1.                                                         |

The **already-built** row in that table is the largest single correction this pass makes. A July 2026 triage pass named a top ten for immediate ingestion; nine of those ten have since been built, along with roughly a dozen other rows it ranked highly. Only Lee 2014 remains outstanding, and it is row 4 here. Any list that still treats that top ten as outstanding work is stale.

## 6.1 Four rows were not what their names said ✗

Checking a name against the built list is not sufficient. A dataset can carry a loader, a failed retrieval attempt, or an already-ingested deposit under a different paper’s name. Three rows did one of those each, and a fourth was caught by the script in Sec. 6.2: O’Duibhir 2014’s expression data is the Kemmeren compendium plus PCA transforms, already built, as that loader’s own docstring states. It is the first duplicate here found mechanically rather than by a reader, which is the point of the check.

**Wildenhain 2016 was already built, under the 2015 name.** It is a data descriptor rather than a second experiment, and its own Data Citation 1 is NCBI PubChem BioAssay AID 1159580, which is exactly the accession `torchcell/datasets/scerevisiae/wildenhain2015.py` already ingests. The identity is not a judgment call: the 2016 paper reports 242 sentinel strains, 5,518 compounds and 492,126 interaction tests, and the loader’s own docstring records 242 systematic ORFs and 492,126 datapoints from that AID. The row is removed.

That leaves a citation-fidelity defect in the built dataset worth fixing separately. The data it ingests is the *extended* matrix described by the 2016 release, but it is filed under the 2015 paper, which reports a strictly smaller experiment: 195 sentinels and 4,915 compounds. The loader already half-flags this, noting that the released data covers 242 ORFs rather than the 195-strain panel it quotes. Under the sourcing rule that a value carries the citation it was actually read from, the released matrix should cite the 2016 descriptor, with 2015 as the original subset.

**Lee 2014, row 4, is blocked rather than available.** WS15 already attempted this ingestion. The roadmap records it as awaiting author matrices, which means the Science supplementary tables and the Nislow and Giaever lab portal did not yield per-strain values. Its rank is still earned: 3,356 compounds against the full heterozygous and homozygous collections really is the largest chemical-genetic matrix in existence, and it really is on the YKO genotype axis the built set already uses. What changes is the nature of the work. Row 4 is an author request, not a loader, and the  $1.3 \times 10^7$  instances it contributes to the totals in Sec. 3 should not be counted as reachable until that request succeeds.

**Turco 2023, row 28, already has a loader.** `torchcell/datasets/scerevisiae/yeastphenome.py` consumes the curated YeastPhenome release and retains 49 growth screens, so this row is in flight rather than net new. It is not in the built set, which is why it appears here at all. Two consequences follow. Its de-duplication problem is partly solved already, since the loader excludes the primaries built directly, along with heterozygous-diploid and expression screens. And Lee 2014 is not among its 49 screens, so ingesting YeastPhenome does not backfill row 4.

## 6.2 Overlap is now checked by script, and the check is not yet strong ✗

Three rows turned out to be already held and none was caught by inspection, so the check is now `experiments/database/scripts/check_candidate_overlap.py`, run against the candidate JSON. It applies three keys of increasing strength: the candidate DOI cited in a built loader, an accession token shared with one, and the candidate PMID appearing in an aggregator’s source-PMID column.

It currently reports one overlap, Turco 2023, which Sec. 6.1 already records as a loader in flight; a second, O’Duibhir 2014’s expression compendium, was found and removed during this pass. It reports no overlap with SynLethDB across



3,307 source PMIDs spanning both the lethality and rescue layers. That covers the question of whether the genetic-interaction rows are already re-served: Sharifpoor 2012, Decourty 2021 and Braberg 2020 are all absent. For Sharifpoor the absence is structural rather than a curation gap, since SynLethDB models synthetic lethality and synthetic rescue, both two-loss-of-function events, and synthetic dosage lethality pairs an overexpression with a deletion.

The honest limit is coverage, and the expansion made it worse in relative terms. The script resolves a PMID for 56 of 155 rows and finds a parseable accession in 9. Accession is the strongest key and the one that would have caught Wildenhain 2016, so it is running on about one row in seventeen. Recording each row's deposit identifier is the single highest-value edit available to this list, and it now matters more than it did at 73 rows.

### 6.3 The expansion tranche is less checked than the first 73 ✗

Rows carry a confidence value in the generated JSON. *sourced* means the figures trace to something read this session or to the sourced triage note, and covers 73 rows. *recall* means the row was written from domain knowledge, and covers 82. For a recall row the dataset is real and the description is sound, but the citation details, the exact counts and the accession all need confirming before a loader is written, and a few carry an explicit warning in the accession column where a collision or a species question is open.

The two tranches are not interleaved by accident: ranking put several recall rows at the very top, because de Boer 2020 and Boocock 2025 genuinely are the largest things here. That is the right ranking and the wrong reading if the confidence flag is ignored. Treat the top of the table as a claim to verify first, not a queue to start work from.

## 7 Before a loader is written ✗

Every figure in Table 1 is an axis count, not a verified record count, and several are weaker than that. Three qualifications apply, in decreasing order of how much damage they would do if ignored.

**Instance counts marked † are products, not counts.** A screen reported as 4,800 strains against 3 conditions is entered as 14,400 instances. Real screens drop strains per condition, so each such figure is an upper bound. The ones marked ‡ are weaker still: order-of-magnitude estimates for libraries whose exact size was not confirmed this pass. Neither should be quoted as a dataset size without opening the source.

**Phenotype dimensionality is estimated where no fixed panel is published.** Ranking uses instances times dimensionality, and dimensionality is exact for a stated panel such as Ambroset 2014 at 74 metabolites, but estimated for open-ended ones. Proteome rows are entered at 2,000 proteins per strain and transcriptome rows at 6,000 genes. Those estimates move rank order, so a row whose position looks surprising should have its dimensionality checked first.

**Accessions marked unconfirmed were not fetched.** The accession column separates locations verified live this pass from those taken from a paper's data-availability statement. Confirmed live: GEO GSE123118 with its companion GitHub release for Galardini 2019, SRA PRJNA986287 for Hale 2024, PRIDE PXD048219 for Muenzner 2024, Dryad for Chica 2026, ChemGRID for Wildenhain 2016, and BioProject PRJNA379146 for Cubillos 2017. Everything else is a claim from a publication, and the pattern is well established that a data-availability statement and a reachable file are different things.

Four rows carry a risk specific enough to name. Triikka 2015 is recorded as figure-only, so whether Additional file 1 holds per-strain scores decides whether Triikka is ingestible at all. Chang 2012 and the Gurvitz fatty-acid screen are PDF-table-only and need manual transcription. Sugiyama is a non-peer-reviewed technical report and has the lowest provenance confidence of anything listed; it belongs in the reserve as a source of candidate genes rather than as training data.

**Two rows are not single datasets.** Turco 2023 is a meta-aggregation and Zackrisson 2016 is a platform. Neither reduces to one loader: Turco needs a de-duplication pass against the rest of this table before its net-new content is known, and Scan-o-matic needs a specific released condition panel chosen before it becomes a row rather than a plan.